

Publications Stefan Schmid (chronological)

- [1] Adrian Kuenzler and Stefan Schmid. Communication bias in large language models: A regulatory perspective. In *Communications of the ACM (CACM)*, 2026.
- [2] Vamsi Addanki, Julien Dallot, Leon Kellerhals, Maciej Pacut, and Stefan Schmid. The harmonic policy for online buffer sharing is $(2 + \ln n)$ -competitive: A simple proof. In *SIAM Symposium on Simplicity in Algorithms (SOSA)*, 2026.
- [3] Aleksander Figiel, Andre Nichterlein, and Stefan Schmid. Efficient algorithms for temporal balanced graph partitioning of datacenter workloads. In *SIAM Symposium on Algorithm Engineering and Experiments (ALENEX)*, 2026.
- [4] Chen Avin, Robert Elsaesser, Aleksander Figiel, Darya Melnyk, and Stefan Schmid. Demand-aware small-world networks on clustered demands. In *29th International Conference on Principles of Distributed Systems (OPODIS)*, 2025.
- [5] Matthias Bentert, Esra Ceylan, Valentin Huebner, Stefan Schmid, and Jiri Srba. Fast re-routing in networks: On the complexity of perfect resilience. In *29th International Conference on Principles of Distributed Systems (OPODIS)*, 2025.
- [6] Stanislava Slavova, Naveed Ali Khan, and Stefan Schmid. Trade-offs between fingerprinting and self-organizing maps in ris-based indoor positioning. In *IEEE Access*, 2025.
- [7] Wenkai Dai, Klaus-Tycho Foerster, and Stefan Schmid. Fast rerouting against dynamic failures: 2-resilience via ear-decomposition and planarity. In *29th International Conference on Principles of Distributed Systems (OPODIS)*, 2025.
- [8] Habib Mostafaei, Mohammad Ezzati, Pieter J. L. Cuijpers, Stefan Schmid, Gabor Retvari, and Sem Borst. Exp-pifo: Scalable and efficient programmable packet scheduling. In *21st ACM International Conference on Emerging Networking Experiments and Technologies (CoNEXT)*, 2025.
- [9] Fabian Goettsch, Max Franke, Arash Pourdamghani, Giuseppe Caire, and Stefan Schmid. Optimizing fronthaul quantization for flexible user load in cell-free massive mimo. In *IEEE Globecom Workshop on Resilience in Next-Generation Wireless Communication Networks*, 2025.
- [10] Simon Scherrer, Adrian Perrig, and Stefan Schmid. A control-theoretic perspective on bbr/cubic congestion-control competition. In *IFIP Performance*, 2025.
- [11] Krishnendu Chatterjee, Jan Matyas Kristan, Stefan Schmid, Jakub Svoboda, and Michelle Yeo. Boosting payment channel network liquidity with topology optimization and transaction selection. In *International Symposium on Distributed Computing (DISC)*, 2025.
- [12] Jan Batzner, Volker Stocker, Stefan Schmid, and Gjergji Kasneci. Germanpartiesqa: Benchmarking commercial large language models and ai companions for political alignment and sycophancy. In *8th AAAI/ACM Conference on AI, Ethics, and Society (AIES)*, 2025.
- [13] Jan Batzner, Volker Stocker, Bingjun Tang, Anusha Natarajan, Qinhao Chen, Stefan Schmid, and Gjergji Kasneci. Whose personae? synthetic persona experiments in llm research and pathways to transparency. In *8th AAAI/ACM Conference on AI, Ethics, and Society (AIES)*, 2025.
- [14] Bruno Maria Rene Gonzalez, Peter Gjol Jensen, Jiri Srba, Stefan Schmid, and Martin Zimmermann. Tapaal hyperltl: A tool for checking hyperproperties of petri nets. In *23rd International Symposium on Automated Technology for Verification and Analysis (ATVA)*, 2025.

- [15] Qingyun Ji, Darya Melnyk, Arash Pourdamghani, and Stefan Schmid. Towards demand-aware peer selection with xor-based routing (invited paper). In *27th International Symposium on Stabilization, Safety, and Security of Distributed Systems (SSS)*, 2025.
- [16] Baosen Zhao, Jianan Sun, Xu Zhou, Wanghong Yang, Wenji Du, Fukang Chen, Yongmao Ren, and Stefan Schmid. Efficient cross-datacenter congestion control with fast control loops. In *54th International Conference on Parallel Processing (ICPP)*, 2025.
- [17] Habib Mostafaei, Maciej Pacut, and Stefan Schmid. Rifo: Pushing the efficiency of programmable packet schedulers. In *IEEE/ACM Transactions on Networking (TON)*, 2025.
- [18] Alexandros Papadopoulos, Dimitrios Tyrovolas, Antonios Lalas, Konstantinos Votis, Stefan Schmid, Sotiris Ioannidis, George K. Karagiannidis, and Christos K. Liaskos. On modeling the ris as a resource: Multi-user allocation and efficiency-proportional pricing. In *IEEE Transactions on Network and Service Management (TNSM)*, 2025.
- [19] Maxim Buzdalov, Pavel Martynov, Sergey Pankratov, Vitaly Aksenov, and Stefan Schmid. In the search of optimal tree networks: Hardness and heuristics. In *The Genetic and Evolutionary Computation Conference (GECCO)*, 2025.
- [20] Marco Reisacher, Ann-Kristin Bergmann, Andreas Blenk, and Stefan Schmid. Breaking the vision: Assessing and mitigating the impact of video artifacts on ml models in industrial use cases. In *IEEE 23rd International Conference on Industrial Informatics (INDIN)*, 2025.
- [21] Melanie Cambus, Darya Melnyk, Tijana Milentijevic, and Stefan Schmid. Approximate agreement algorithms for byzantine collaborative learning. In *37th ACM Symposium on Parallelism in Algorithms and Architectures (SPAA)*, 2025.
- [22] Harald Raecke, Stefan Schmid, and Ruslan Zabrodin. Tight bounds for online balanced partitioning in the generalized learning model. In *37th ACM Symposium on Parallelism in Algorithms and Architectures (SPAA)*, 2025.
- [23] Jonas Koeppeler, Toke Hoiland-Jorgensen, and Stefan Schmid. Have your cake and eat it too: Scaling software rate limiting across cpu cores. In *IEEE International Symposium on Local and Metropolitan Area Networks (LANMAN)*, 2025.
- [24] Antoine El-Hayek, Robert Elsässer, and Stefan Schmid. An almost tight lower bound for plurality consensus with undecided state dynamics in the population protocol model. In *ACM Symposium on Principles of Distributed Computing (PODC)*, 2025.
- [25] Krishnendu Chatterjee, Seth Gilbert, Stefan Schmid, Jakub Svoboda, and Michelle Yeo. When is liquid democracy possible? on the manipulation of variance. In *ACM Symposium on Principles of Distributed Computing (PODC)*, 2025.
- [26] Tom-Lukas Breitkopf, Julien Dallot, Antoine El-Hayek, and Stefan Schmid. Brief announcement: Minimizing energy solves relative majority with a cubic number of states in population protocols. In *ACM Symposium on Principles of Distributed Computing (PODC)*, 2025.
- [27] Chen Avin and Stefan Schmid. Revolutionizing datacenter networks via reconfigurable topologies. In *Communications of the ACM (CACM)*, 2025.
- [28] Melanie Cambus, Darya Melnyk, Tijana Milentijevic, and Stefan Schmid. Coordinate-wise median in byzantine federated learning. In *ACM AsiaCCS Workshop on Secure and Efficient Federated Learning (FL-AsiaCCS)*, 2025.

- [29] Aleksander Figiel, Darya Melnyk, Tijana Milentijevic, and Stefan Schmid. Distributed construction of demand-aware datacenter networks. In *39th IEEE International Parallel and Distributed Processing Symposium (IPDPS)*, 2025.
- [30] Burak Arda Okutan, Yvonne-Anne Pignolet, and Stefan Schmid. Democracy for daos: An empirical study of decentralized governance and dynamics. In *IEEE International Conference on Blockchain and Cryptocurrency (ICBC)*, 2025.
- [31] Kexin Liu, Zhaochen Zhang, Chang Liu, Yizhi Wang, Qingyue Wang, Vamsi Addanki, Stefan Schmid, Wei Chen, Xiaoliang Wang, Jiaqi Zheng, Wenhao Sun, Tao Wu, Ke Meng, Fei Chen, Weiguang Wang, Wanchun Dou, Guihai Chen, and Chen Tian. Pyrrha: Congestion-root-based flow control to eliminate head-of-line blocking in datacenter. In *22nd USENIX Symposium on Networked Systems Design and Implementation (NSDI)*, 2025.
- [32] Matthias Bentert, Max Franke, Darya Melnyk, Arash Pourdamghani, and Stefan Schmid. Demand-aware multi-source ip-multicast: Minimal congestion via link weight optimization. In *IFIP Networking*, 2025.
- [33] Wenkai Dai, Klaus-Tycho Foerster, and Stefan Schmid. On the resilience of fast failover routing against dynamic link failures. In *IFIP Networking*, 2025.
- [34] Nicolas Schnepf, Remi Badonnel, Damien Saucez, Stefan Schmid, and Jiri Srba. Eagle: Vulnerability and congestion aware software update synthesis in softwarized networks. In *IEEE/IFIP Network Operations and Management Symposium (NOMS)*, 2025.
- [35] Cao Vien Phung, Max Franke, Ehsan Tohidi, June Heinemann, Andre Drummond, Stefan Schmid, Slawomir Stanczak, and Admela Jukan. Reliability and qos assurance in ris-assisted indoor networks. In *International Conference On The Design Of Reliable Communication Networks (DRCN)*, 2025.
- [36] Lukas Aumayr, Zeta Avarikioti, Iosif Salem, Stefan Schmid, and Michelle Yeo. X-transfer: Enabling and optimizing cross-pcn transactions. In *29th International Conference on Financial Cryptography and Data Security (FC)*, 2025.
- [37] Jonas Koeppeler, Toke Hoiland-Jorgensen, and Stefan Schmid. mq-cake: Scaling software rate limiting across cpu cores. In *The Technical Conference on Linux Networking (Netdev 0x19)*, 2025.
- [38] Aleksander Figiel, Darya Melnyk, Andre Nichterlein, Arash Pourdamghani, and Stefan Schmid. Spiderdan: Matching augmentation in demand-aware networks. In *SIAM Symposium on Algorithm Engineering and Experiments (ALENEX)*, 2025.
- [39] Dor Cohen, Yvonne-Anne Pignolet, Ognjen Maric, and Stefan Schmid. Internet computer as a data availability layer (poster paper). In *IEEE International Conference on Blockchain and Cryptocurrency (ICBC)*, 2025.
- [40] Aleksander Figiel, Janne H. Korhonen, Neil Olver, and Stefan Schmid. Efficient algorithms for demand-aware networks and a connection to virtual network embedding. In *28th International Conference on Principles of Distributed Systems (OPODIS)*, 2024.
- [41] Arash Pourdamghani, Chen Avin, Robert Sama, Maryam Shiran, and Stefan Schmid. Hash and adjust: Competitive demand-aware consistent hashing. In *28th International Conference on Principles of Distributed Systems (OPODIS)*, 2024.
- [42] Joao Paulo Bezerra, Veronika Anikina, Petr Kuznetsov, Liron Schiff, and Stefan Schmid. Dynamic probabilistic reliable broadcast. In *28th International Conference on Principles of Distributed Systems (OPODIS)*, 2024.

- [43] Sarah-Michelle Hammer, Vamsi Addanki, Max Franke, and Stefan Schmid. Starlink performance through the edge router lens. In *2nd ACM MobiCom Workshop on LEO Networking and Communication (LEO-NET)*, 2024.
- [44] Antoine El-Hayek, Monika Henzinger, and Stefan Schmid. Broadcast and consensus in stochastic dynamic networks with byzantine nodes and adversarial edges. In *International Symposium on Distributed Computing (DISC)*, 2024.
- [45] Alexandros Ioannis Papadopoulos, Antonios Lalas, Konstantinos Votis, Stefan Schmid, and Christos Liaskos. Ris as a network resource: Pricing and user multiplexing algorithms. In *International Conference on 6G Networking (6GNet)*, 2024.
- [46] Max Franke, Jake Holland, and Stefan Schmid. Mcquic - a multicast extension for quic. In *22nd International Symposium on Network Computing and Applications (NCA)*, 2024.
- [47] Caio Caldeira, Otavio Augusto de Oliveira Souza, Olga Goussevskaia, and Stefan Schmid. Optical self-adjusting data center networks in the scalable matching model. In *IEEE/ACM Transactions on Cloud Computing (TCC)*, 2024.
- [48] Milad Miri, Stefan Schmid, and Habib Mostafaei. A demand-aware networked system using telemetry and ml with reactnet. In *22nd International Symposium on Network Computing and Applications (NCA)*, 2024.
- [49] Zeta Avarikioti, Stefan Schmid, and Samarth Tawari. Musketeer: Incentive-compatible rebalancing for payment channel networks. In *Advances in Financial Technologies (AFT)*, 2024.
- [50] Evgeniy Feder, Anton Paramonov, Pavel Mavrin, Iosif Salem, Vitaly Aksenov, and Stefan Schmid. Toward self-adjusting k-ary search tree networks. In *European Symposium on Algorithms (ESA)*, 2024.
- [51] Vamsi Addanki, Wei Bai, Stefan Schmid, and Maria Apostolaki. Reverie: Low pass filter-based switch buffer sharing for datacenters with rdma and tcp traffic. In *21st USENIX Symposium on Networked Systems Design and Implementation (NSDI)*, 2024.
- [52] Vamsi Addanki, Maciej Pacut, and Stefan Schmid. Credence: Augmenting datacenter switch buffer sharing with ml predictions. In *21st USENIX Symposium on Networked Systems Design and Implementation (NSDI)*, 2024.
- [53] Csaba Gyorgyi, Kim G. Larsen, Stefan Schmid, and Jiri Srba. Syrep: Efficient synthesis and repair of fast re-route forwarding tables for resilient networks. In *54th IEEE/IFIP International Conference on Dependable Systems and Networks (DSN)*, 2024.
- [54] Julien Dallot, Maciej Pacut, Marcin Bienkowski, Darya Melnyk, and Stefan Schmid. Learning minimum linear arrangement of cliques and lines. In *44th IEEE International Conference on Distributed Computing Systems (ICDCS)*, 2024.
- [55] Stefan Schmid and Dmitry Shestakov. Invited paper: Blockchain governance and liquid democracy – quantifying decentralization in bitcoin and internet computer. In *ApPLIED Workshop at ACM PODC*, 2024.
- [56] Juan Vanerio, Huiran Liu, Qi Zhang, and Stefan Schmid. Baiji: Domain planning for cdns under the 95th percentile billing model. In *IFIP Networking*, 2024.
- [57] Lukas Aumayr, Esra Ceylan, Yannik Kopyciok, Matteo Maffei, Pedro Moreno-Sanchez, Iosif Salem, and Stefan Schmid. Optimizing virtual payment channel establishment in the face of on-path adversaries. In *IFIP Networking*, 2024.

- [58] Radu Vintan, Harald Raecke, and Stefan Schmid. Fast algorithms for loop-free network updates using linear programming and local search. In *IEEE INFOCOM*, 2024.
- [59] Wenkai Dai, Mike Dinitz, Klaus-Tycho Foerster, Long Luo, and Stefan Schmid. Approximation algorithms for minimizing congestion in demand-aware networks. In *IEEE INFOCOM*, 2024.
- [60] Julien Dallot, Amirmehdi Jafari Fesharaki, Maciej Pacut, and Stefan Schmid. Dependency-aware online caching. In *IEEE INFOCOM*, 2024.
- [61] Csaba Gyorgyi, Kim G. Larsen, Stefan Schmid, and Jiri Srba. Syper: Synthesis of perfectly resilient local fast rerouting rules for highly dependable networks. In *IEEE INFOCOM*, 2024.
- [62] Esra Ceylan, Jakub Svoboda, Krishnendu Chatterjee, and Stefan Schmid. Congestion-free rerouting of network flows: Hardness and an fpt algorithm. In *IEEE/IFIP Network Operations and Management Symposium (NOMS), Mini-Conference*, 2024.
- [63] Olivier Michaelis, Stefan Schmid, and Habib Mostafaei. L3: Latency-aware load balancing in multi-cluster service mesh. In *25th ACM/IFIP International Middleware Conference (Middleware)*, 2024.
- [64] Yvonne-Anne Pigolet, Stefan Schmid, and Arne Seelisch. Gender-specific homophily on instagram and implications on information spread. In *Nature Scientific Reports*, 2024.
- [65] Ehsan Tohidi, Max Franke, Cao Vien Phung, Naveed Ali Khan Kaim Khani, Andre C. Drummond, Stefan Schmid, Admela Jukan, and Slawomir Stanczak. Reliability assurance in ris-assisted 6g campus networks. In *International Conference On The Design Of Reliable Communication Networks (DRCN)*, 2024.
- [66] Marcin Bienkowski and Stefan Schmid. A subquadratic bound for online bisection. In *41st International Symposium on Theoretical Aspects of Computer Science (STACS)*, 2024.
- [67] Philipp Zabka, Klaus-Tycho Foerster, Christian Decker, and Stefan Schmid. A centrality analysis of the lightning network. In *Telecommunications Policy (JTPO), Elsevier, Special issue: Economic, social and political impact of blockchain*, 2024.
- [68] Johannes Gruendler, Darya Melnyk, Arash Pourdamghani, and Stefan Schmid. Decentpeer: A self-incentivised and inclusive decentralized peer review system (poster paper). In *IEEE International Conference on Blockchain and Cryptocurrency (ICBC)*, 2024.
- [69] Stefan Schmid, Jakub Svoboda, and Michelle Yeo. Weighted packet selection for rechargeable links in cryptocurrency networks: Complexity and approximation. In *Theoretical Computer Science (TCS), Elsevier*, 2024.
- [70] Sanja Bonic, Janos Bonic, and Stefan Schmid. Broomrocket: Open source text-to-3d algorithm for 3d object placement. In *ACM Games: Research and Practice*, 2024.
- [71] Hugo Rincon Galeana, Ami Paz, Stefan Schmid, Ulrich Schmid, and Kyrill Winkler. The time complexity of consensus under oblivious message adversaries. In *Theoretical Computer Science (TCS), Elsevier*, 2024.
- [72] Naveed Ali Kaim Khani and Stefan Schmid. Ai-ran in 6g networks state-of-the-art and challenges. In *IEEE Open Journal of the Communications Society*, 2024.
- [73] Juan Vanerio, Lily Huegerich, and Stefan Schmid. Tero: Offloading cdn traffic to massively distributed devices. In *25th International Conference on Distributed Computing and Networking (ICDCN)*, 2024.
- [74] Juho Hirvonen, Laura Schmid, Krishnendu Chatterjee, and Stefan Schmid. On the convergence time in graphical games: A locality-sensitive approach. In *27th International Conference on Principles of Distributed Systems (OPODIS)*, 2023.

- [75] Marcin Bienkowski, David Fuchssteiner, and Stefan Schmid. Optimizing reconfigurable optical data-centers: The power of randomization. In *International Conference for High Performance Computing, Networking, Storage, and Analysis (SC)*, aka *ACM/IEEE Supercomputing*, 2023.
- [76] Joern-Thorben Hinz, Vamsi Addanki, Csaba Gyorgyi, Theo Jepsen, and Stefan Schmid. Tcp’s third eye: Leveraging ebpf for telemetry-powered congestion control. In *SIGCOMM Workshop on eBPF and Kernel Extensions (eBPF)*, 2023.
- [77] Yuval Emek, Yuval Gil, Maciej Pacut, and Stefan Schmid. Online algorithms with randomly infused advice. In *European Symposium on Algorithms (ESA)*, 2023.
- [78] Vincent Happersberger, Frank-Walter Jaekel, Thomas Knothe, Yvonne-Anne Pignolet, and Stefan Schmid. Comparison of ethereum smart contract analysis and verification methods. In *7th International Workshop on Cryptocurrencies and Blockchain Technology (CBT)*, 2023.
- [79] Anya Chaturvedi, Chandra Chekuri, Andrea Richa, Matthias Rost, Stefan Schmid, and Jamison Weber. Improved throughput for all-or-nothing multicommodity flows with arbitrary demands. In *IEEE/ACM Transactions on Networking (TON)*, 2023.
- [80] Efim Zhitomirskiy, Stefan Schmid, and Martin Walther. Tokenizing assets with dividend payouts - a legally compliant and flexible design. In *Digital Finance*, Springer, 2023.
- [81] Kim Guldstrand Larsen, Anders Mariegaard, Stefan Schmid, and Jiri Srba. Allsynth: A bdd-based approach for network update synthesis. In *Science of Computer Programming (SCICO)*, 2023.
- [82] Christos Liaskos, Kostas Katsalis, Joan Triay, and Stefan Schmid. Resource management for programmable metasurfaces: Concept, prospects and challenges. In *IEEE Communications Magazine (COMMAG)*, 2023.
- [83] Christoph Lenzen, Moti Medina, Mehrdad Saberi, and Stefan Schmid. Robust routing made easy: Reinforcing networks against non-benign faults. In *IEEE/ACM Transactions on Networking (TON)*, 2023.
- [84] Vamsi Addanki, Chen Avin, and Stefan Schmid. Mars: Near-optimal throughput with shallow buffers in reconfigurable datacenter networks. In *ACM SIGMETRICS*, 2023.
- [85] Johannes Zerwas, Csaba Gyorgyi, Andreas Blenk, Stefan Schmid, and Chen Avin. Duo: A high-throughput reconfigurable datacenter network using local routing and control. In *ACM SIGMETRICS*, 2023.
- [86] Csaba Gyorgyi, Sandor Laki, and Stefan Schmid. Toward highly reliable programmable data planes: Verification of p4 code generation. In *9th IEEE International Conference on Network Softwarization (NetSoft)*, 2023.
- [87] Chen Avin, Ingo van Duijn, Maciej Pacut, and Stefan Schmid. Self-adjusting grid networks. In *Elsevier Information and Computation*, 2023.
- [88] Harald Raecke, Stefan Schmid, and Ruslan Zabrodin. Polylog-competitive algorithms for dynamic balanced graph partitioning for ring demands. In *35th ACM Symposium on Parallelism in Algorithms and Architectures (SPAA)*, 2023.
- [89] Wenkai Dai, Klaus-Tycho Foerster, and Stefan Schmid. A tight characterization of fast failover routing: Resiliency to two link failures is possible. In *35th ACM Symposium on Parallelism in Algorithms and Architectures (SPAA)*, 2023.
- [90] Jerome De Boeck, Bernard Fortz, and Stefan Schmid. The case for stochastic online segment routing under demand uncertainty. In *IFIP Networking*, 2023.

- [91] Stefan Schmid, Jakub Svoboda, and Michelle Yeo. Weighted packet selection for rechargeable links in cryptocurrency networks: Complexity and approximation. In *30th International Colloquium on Structural Information and Communication Complexity (SIROCCO)*, 2023.
- [92] Vitaly Aksenov, Anton Paramonov, Iosif Salem, and Stefan Schmid. Self-adjusting linear networks with ladder demand graph. In *30th International Colloquium on Structural Information and Communication Complexity (SIROCCO)*, 2023.
- [93] Klaus-Tycho Foerster, Thibault Marette, Stefan Neumann, Claudia Plant, Ylli Sadikaj, Stefan Schmid, and Yllka Velaj. Analyzing the communication clusters in datacenters. In *The Web Conference (WWW)*, 2023.
- [94] Mahsa Bastankhah, Krishnendu Chatterjee, Mohammad Ali Maddah-Ali, Stefan Schmid, Jakub Svoboda, and Michelle Yeo. R2: Boosting liquidity in payment channel networks with online admission control (combining on-chain recharging with off-chain rebalancing). In *Financial Cryptography and Data Security (FC)*, 2023.
- [95] Monika Henzinger, Stefan Neumann, Harald Raecke, and Stefan Schmid. Dynamic maintenance of monotone dynamic programs and applications. In *40th International Symposium on Theoretical Aspects of Computer Science (STACS)*, 2023.
- [96] Arash Pourdamghani, Chen Avin, Robert Sama, and Stefan Schmid. Seedtree: A dynamically optimal and local self-adjusting tree. In *IEEE INFOCOM*, 2023.
- [97] Vamsi Addanki, Maciej Pacut, Arash Pourdamghani, Gabor Retvari, Stefan Schmid, and Juan Vanerio. Self-adjusting partially ordered lists. In *IEEE INFOCOM*, 2023.
- [98] Kathrin Hanauer, Monika Henzinger, Lara Ost, and Stefan Schmid. Dynamic demand-aware link scheduling for reconfigurable datacenters. In *IEEE INFOCOM*, 2023.
- [99] Caio Caldeira, Otavio Augusto de Oliveira Souza, Olga Goussevskaia, and Stefan Schmid. Opticnet: Self-adjusting networks for tor-matching-tor optical switching architectures. In *IEEE INFOCOM*, 2023.
- [100] Apoorv Shukla, Kevin Hudemann, Zsolt Vagi, Lily Hugerich, Georgios Smaragdakis, Artur Hecker, Stefan Schmid, and Anja Feldmann. Runtime verification for programmable switches. In *IEEE/ACM Transactions on Networking (TON)*, 2023.
- [101] Alkida Balliu, Janne H. Korhonen, Fabian Kuhn, Henrik Lievonen, Dennis Olivetti, Shreyas Pai, Ami Paz, Joel Rybicki, Stefan Schmid, Jan Studený, Jukka Suomela, and Jara Uitto. Sinkless orientation made simple. In *SIAM Symposium on Simplicity in Algorithm (SOSA)*, 2023.
- [102] Hugo Rincon Galeana, Ami Paz, Stefan Schmid, Ulrich Schmid, and Kyrill Winkler. The time complexity of consensus under oblivious message adversaries. In *14th Innovations in Theoretical Computer Science (ITCS)*, 2023.
- [103] Antoine El-Hayek, Monika Henzinger, and Stefan Schmid. Asymptotically tight bounds on the time complexity of broadcast and its variants in dynamic networks. In *14th Innovations in Theoretical Computer Science (ITCS)*, 2023.
- [104] Elif Bozkaya, Berk Canberk, and Stefan Schmid. Digital twin-empowered resource allocation for 6g-enabled massive iot. In *Workshop on the Evolution of Digital Twin Paradigm in Wireless Communications*, 2023.
- [105] Wenkai Dai, Klaus-Tycho Foerster, David Fuchssteiner, and Stefan Schmid. Load-optimization in reconfigurable networks: Algorithms and complexity of flow routing. In *Transactions on Modeling and Performance Evaluation of Computing Systems (TOMPECS)*, 2023.

- [106] Csaba Györgyi, Sandor Laki, and Stefan Schmid. P4rrot: Generating p4 code for the application layer. In *ACM SIGCOMM Computer Communication Review (CCR)*, 2023.
- [107] C. Liaskos, A. Tsioliaridou, K. Georgopoulos, G. Morianos, S. Ioannidis, I. Salem, D. Manassis, S. Schmid, D. Tyrovolas, S. A. Tegos, P.-V. Mekikis, P. D. Diamantoulakis, A. Pitilakis, N. Kantartzis, G. K. Karagiannidis, A. Tasolamprou, O. Tsilipakos, M. Kafesaki, I.F. Akyildiz, A. Pitsillides, M. Pateraki, M. Vakalellis, and I. Spais. Xr-rf imaging enabled by software-defined metasurfaces and machine learning: Foundational vision, technologies and challenges. In *IEEE Access*, 2023.
- [108] Neta Rozen Schiff, Klaus-Tycho Foerster, Stefan Schmid, and David Hay. Chopin: Combining distributed and centralized schedulers for self-adjusting datacenter networks. In *25th International Conference on Principles of Distributed Systems (OPODIS)*, 2022.
- [109] Stefan Schmid, Morten Konggaard Schou, Jiri Srba, and Juan Vanerio. R-mpls: Recursive protection for highly dependable mpls networks. In *18th ACM International Conference on Emerging Networking Experiments and Technologies (CoNEXT)*, 2022.
- [110] Juan Vanerio, Stefan Schmid, Morten Schou, and Jiri Srba. Mpls-kit: An mpls data plane toolkit. In *IEEE Global Internet Symposium (GI)*, 2022.
- [111] Nicklas Johansen, Lasse Kaer, Andreas Madsen, Kristian Nielsen, Stefan Schmid, Jiri Srba, and Rasmus Tollund. Fbr: Dynamic memory-aware fast rerouting. In *IEEE Global Internet Symposium (GI)*, 2022.
- [112] Yvonne-Anne Pignolet and Stefan Schmid. Die technologischen ansatze und herausforderungen des web3 und metaverse. In *Telemedicus – Recht der Informationsgesellschaft, Tagungsband zur Sommerkonferenz, Deutscher Fachverlag*, 2022.
- [113] Peter Gjol Jensen, Stefan Schmid, Morten Konggaard Schou, and Jiri Srba. Pdaaal: A library for reachability analysis of weighted pushdown systems. In *20th International Symposium on Automated Technology for Verification and Analysis (ATVA)*, 2022.
- [114] Simon Scherrer, Markus Legner, Adrian Perrig, and Stefan Schmid. Model-based insights on the performance, fairness, and stability of bbr. In *ACM Internet Measurement Conference (IMC)*, 2022.
- [115] Maciej Pacut, Mahmoud Parham, Joel Rybicki, Stefan Schmid, Jukka Suomela, and Aleksandr Tereshchenko. Brief announcement: Temporal locality in online algorithms. In *International Symposium on Distributed Computing (DISC)*, 2022.
- [116] Wenkai Dai, Michael Dinitz, Klaus-Tycho Foerster, and Stefan Schmid. Brief announcement: Minimizing congestion in hybrid demand-aware network topologies. In *International Symposium on Distributed Computing (DISC)*, 2022.
- [117] Monika Henzinger, Ami Paz, Arash Pourdamghani, and Stefan Schmid. The augmentation-speed tradeoff for consistent network updates. In *ACM SIGCOMM Symposium on SDN Research (SOSR)*, 2022.
- [118] Samarth Tiwari, Michelle Yeo, Zeta Avarikioti, Iosif Salem, Krzysztof Pietrzak, and Stefan Schmid. Wiser: Increasing throughput in payment channel networks with transaction aggregation. In *Proc. 4th ACM Conference on Advances in Financial Technologies (AFT)*, 2022.
- [119] Vamsi Addanki, Maria Apostolaki, Manya Ghobadi, Stefan Schmid, and Laurent Vanbever. Abm: Active buffer management in datacenters. In *Proc. ACM SIGCOMM*, 2022.
- [120] Harald Raecke, Stefan Schmid, and Ruslan Zabrodin. Approximate dynamic balanced graph partitioning. In *Proc. 34th ACM Symposium on Parallelism in Algorithms and Architectures (SPAA)*, 2022.

- [121] Antoine El-Hayek, Monika Henzinger, and Stefan Schmid. Brief announcement: Broadcasting time in dynamic rooted trees is linear. In *Proc. ACM Symposium on Principles of Distributed Computing (PODC)*, 2022.
- [122] Gregor Bankhamer, Robert Elsaesser, and Stefan Schmid. Local fast rerouting with low congestion: A randomized approach. In *IEEE/ACM Transactions on Networking (TON)*, 2022.
- [123] Kim Guldstrand Larsen, Anders Mariegaard, Stefan Schmid, and Jiri Srba. Allsynth: Transiently correct network update synthesis accounting for operator preferences. In *Proc. 16th International Symposium on Theoretical Aspects of Software Engineerin (TASE)*, 2022.
- [124] Chen Avin, Kaushik Mondal, and Stefan Schmi. Push-down trees: Optimal self-adjusting complete trees. In *IEEE/ACM Transactions on Networking (TON)*, 2022.
- [125] Chen Avin, Marcin Bienkowski, Iosif Salem, Robert Sama, Stefan Schmid, and Paweł Schmidt. Deterministic self-adjusting tree networks using rotor walks. In *Proc. 42nd IEEE International Conference on Distributed Computing Systems (ICDCS)*, 2022.
- [126] Stefan Schmid, Bernhard Schrenk, and Alvaro Torralba. Netstack: A game approach to synthesizing consistent network updates. In *Proc. IFIP Networking*, 2022.
- [127] Philipp Zabka, Klaus-Tycho Foerster, Stefan Schmid, and Christian Decker. Empirical evaluation of nodes and channels of the lightning network. In *Pervasive and Mobile Computing (PMC)*, 2022.
- [128] Klaus-Tycho Foerster, Juho Hirvonen, Yvonne-Anne Pignolet, Stefan Schmid, and Gilles Tredan. On the price of locality in static fast rerouting. In *Proc. 52nd IEEE/IFIP International Conference on Dependable Systems and Networks (DSN)*, 2022.
- [129] Otavio Augusto de Oliveira Souza, Olga Goussevskaia, and Stefan Schmid. Cbnet: Demand-aware tree topologies for reconfigurable datacenter networks. In *Journal Computer Networks (COMNET)*, 2022.
- [130] Long Luo, Klaus-Tycho Foerster, Stefan Schmid, and Hongfang Yu. Optimizing multicast flows in high-bandwidth reconfigurable datacenter networks. In *Journal of Network and Computer Applications (JNCA)*, 2022.
- [131] Chen Griner, Johannes Zerwas, Andreas Blenk, Manya Ghobadi, Stefan Schmid, and Chen Avin. Cerberus: The power of choices in datacenter topology design (a throughput perspective). In *Proc. of the ACM on Measurement and Analysis of Computing Systems (POMACS)*, 2022.
- [132] Pieter Cuijpers, Stefan Schmid, Nicolas Schnepf, and Jiri Srba. The hazard value: A quantitative network connectivity measure accounting for failures. In *Proc. 52nd IEEE/IFIP International Conference on Dependable Systems and Networks (DSN)*, 2022.
- [133] Chen Avin, Kaushik Mondal, and Stefan Schmi. Demand-aware network design with minimal congestion and route lengths. In *IEEE/ACM Transactions on Networking (TON)*, 2022.
- [134] Vamsi Addanki, Oliver Michel, and Stefan Schmid. Powertcp: Pushing the performance limits of datacenter networks. In *Proc. 19th USENIX Symposium on Networked Systems Design and Implementation (NSDI)*, 2022.
- [135] Kathrin Hanauer, Monika Henzinger, Stefan Schmid, and Jonathan Trummer. Fast and heavy disjoint weighted matchings for demand-aware datacenter topologies. In *Proc. IEEE Conference on Computer Communications (INFOCOM)*, 2022.
- [136] Evgeniy Feder, Ichha Rathod, Punit Shyamsukha, Robert Sama, Vitaly Aksenov, Iosif Salem, and Stefan Schmid. Lazy self-adjusting bounded-degree networks for the matching model. In *Proc. IEEE Conference on Computer Communications (INFOCOM)*, 2022.

- [137] Zeta Avarikioti, Krzysztof Pietrzak, Iosif Salem, Stefan Schmid, Samarth Tiwari, and Michelle Yeo. Hide and seek: Privacy-preserving rebalancing on payment channel networks. In *Proc. Financial Cryptography and Data Security (FC)*, 2022.
- [138] Philipp Zabka, Klaus-Tycho Foerster, Christian Decker, and Stefan Schmid. Short paper: A centrality analysis of the lightning network. In *Proc. Financial Cryptography and Data Security (FC)*, 2022.
- [139] Utz Nisslmüller, Klaus-Tycho Foerster, Stefan Schmid, and Christian Decker. Inferring sensitive information in cryptocurrency off-chain networks using probing and timing attacks. In *Information Systems Security and Privacy, Communications in Computer and Information Science (CCIS)*, 2022.
- [140] Marco Canini, Iosif Salem, Liron Schiff, Elad Michael Schiller, and Stefan Schmid. Renaissance: A self-stabilizing distributed sdn control plane using in-band communications. In *Journal of Computer and System Sciences (JCSS)*, 2022.
- [141] Peter Gjol Jensen, Jesper Stenbjerg Jensen, Troels Beck Krogh, Jonas Sand Madsen, Stefan Schmid, Jiri Srba, Marc Tom Thorgersen, and Ingo van Duijn. Automata theoretic approach to verification of mpls networks under link failures. In *IEEE/ACM Transactions on Networking (TON)*, 2021.
- [142] Maciej Pacut, Wenkai Dai, Alexandre Labbe, Klaus-Tycho Foerster, and Stefan Schmid. Improved scalability of demand-aware datacenter topologies with minimal route lengths and congestion. In *Performance Evaluation (PEVA)*, 2021.
- [143] Chen Griner, Stefan Schmid, and Chen Avin. Cachenet: Leveraging the principle of locality in reconfigurable network design. In *Journal Computer Networks (COMNET)*, 2022.
- [144] Chen Griner, Johannes Zerwas, Andreas Blenk, Manya Ghobadi, Stefan Schmid, and Chen Avin. Cerberus: The power of choices in datacenter topology design (a throughput perspective). In *Proc. ACM SIGMETRICS*, 2022.
- [145] Marcelo Abranches, Oliver Michel, Eric Keller, and Stefan Schmid. Efficient network monitoring applications in the kernel with ebpf and xdp. In *Proc. 7th IEEE Conference on Network Functions Virtualization and Software-Defined Networking (NFV-SDN)*, 2021.
- [146] Bruna Peres, Otavio Augusto de Oliveira Souza, Olga Goussevskaia, Chen Avin, and Stefan Schmid. Distributed self-adjusting tree networks. In *IEEE Transactions on Cloud Computing (TCC)*, 2021.
- [147] Mahmoud Parham, Thomas Fenz, Nikolaus Suess, Klaus-Tycho Foerster, and Stefan Schmid. Traffic engineering with joint link weight and segment optimization. In *Proc. 17th ACM International Conference on emerging Networking EXperiments and Technologies (CoNEXT)*, 2021.
- [148] Johannes Zerwas, Kaan Aykurt, Stefan Schmid, and Andreas Blenk. Network traffic characteristics of machine learning frameworks under the microscope. In *17th International Conference on Network and Service Management (CNSM)*, 2021.
- [149] Johannes Zerwas, Ingmar Poesse, Stefan Schmid, and Andreas Blenk. On the benefits of joint optimization of reconfigurable cdn-isp infrastructure. In *IEEE Transactions on Network and Service Management (TNSM)*, 2021.
- [150] Anya Chaturvedi, Chandra Chekuri, Andrea Richa, Matthias Rost, Stefan Schmid, and Jamison Weber. Improved throughput for all-or-nothing multicommodity flows with arbitrary demands. In *39th International Symposium on Computer Performance, Modeling, Measurements and Evaluation (PERFORMANCE)*, 2021.
- [151] Simon Scherrer, Markus Legner, Adrian Perrig, and Stefan Schmid. An axiomatic perspective on the performance effects of end-host path selection. In *39th International Symposium on Computer Performance, Modeling, Measurements and Evaluation (PERFORMANCE)*, 2021.

- [152] Maciej Pacut, Wenkai Dai, Alexandre Labbe, Klaus-Tycho Foerster, and Stefan Schmid. Improved scalability of demand-aware datacenter topologies with minimal route lengths and congestion. In *39th International Symposium on Computer Performance, Modeling, Measurements and Evaluation (PERFORMANCE)*, 2021.
- [153] Gregor Bankhamer, Robert Elsässer, and Stefan Schmid. Randomized local fast rerouting for data-center networks with almost optimal congestion. In *International Symposium on Distributed Computing (DISC)*, 2021.
- [154] Janne H. Korhonen, Ami Paz, Joel Rybicki, Stefan Schmid, and Jukka Suomela. Brief announcement: Sinkless orientation is hard also in the supported local model. In *International Symposium on Distributed Computing (DISC)*, 2021.
- [155] Peter Gjol Jensen, Stefan Schmid, Morten Konggaard Schou, Jiri Srba, Juan Vanerio, and Ingo van Duijn. Faster pushdown reachability analysis with applications in network verification. In *19th International Symposium on Automated Technology for Verification and Analysis (ATVA)*, 2021.
- [156] Yvonne-Anne Pignolet, Stefan Schmid, and Gilles Tredan. On the implications of routing models on network optimization. In *IEEE Transactions on Network and Service Management (TNSM)*, 2021.
- [157] Kashyap Thimmaraju, Liron Schiff, and Stefan Schmid. Preacher: Network policy checker for adversarial environments. In *IEEE/ACM Transactions on Networking*, 2021.
- [158] Janardhan Kulkarni, Stefan Schmid, and Pawel Schmidt. Scheduling opportunistic links in two-tiered reconfigurable datacenters. In *33rd ACM Symposium on Parallelism in Algorithms and Architectures (SPAA)*, 2021.
- [159] Aleksander Figiel, Leon Kellerhals, Rolf Niedermeier, Matthias Rost, Stefan Schmid, and Philipp Zschoche. Optimal virtual network embeddings for tree topologies. In *33rd ACM Symposium on Parallelism in Algorithms and Architectures (SPAA)*, 2021.
- [160] Evgeniy Feder, Ichha Rathod, Punit Shyamsukha, Robert Sama, Vitaly Aksenov, Iosif Salem, and Stefan Schmid. Brief announcement: Toward self-adjusting networks for the matching model. In *33rd ACM Symposium on Parallelism in Algorithms and Architectures (SPAA)*, 2021.
- [161] Matthew Nance Hall, Klaus-Tycho Foerster, Stefan Schmid, and Ramakrishnan Durairajan. A survey of reconfigurable optical networks. In *Optical Switching and Networking (OSN)*, Elsevier, 2021.
- [162] Simon Scherrer, Markus Legner, Adrian Perrig, and Stefan Schmid. Enabling novel interconnection agreements with path-aware network architectures. In *Proc. 51st IEEE/IFIP International Conference on Dependable Systems and Networks (DSN)*, 2021.
- [163] Krzysztof Pietrzak, Iosif Salem, Stefan Schmid, and Michelle Yeo. Lightpir: Privacy-preserving route discovery for payment channel networks. In *Proc. IFIP Networking*, 2021.
- [164] Monika Henzinger, Ami Paz, and Stefan Schmid. On the complexity of weight-dynamic network algorithms. In *Proc. IFIP Networking*, 2021.
- [165] Esra Ceylan, Klaus-Tycho Foerster, Stefan Schmid, and Katsiaryna Zaitsava. Demand-aware plane spanners of bounded degree. In *Proc. IFIP Networking*, 2021.
- [166] Robin Muenk, Matthias Rost, Harald Raecke, and Stefan Schmid. It’s good to relax: Fast profit approximation for virtual networks with latency constraints. In *Proc. IFIP Networking*, 2021.
- [167] Oliver Michel, Roberto Bifulco, Gabor Retvari, and Stefan Schmid. The programmable data plane: Abstractions, architectures, algorithms, and applications. In *Proc. ACM Computing Surveys (CSUR)*, 2021.

- [168] Stefan Schmid, Nicolas Schnepf, and Jiri Srba. Resilient capacity-aware routing. In *Proc. 27th International Conference on Tools and Algorithms for the Construction and Analysis of Systems (TACAS)*, 2021.
- [169] Klaus-Tycho Foerster, Janne Korhonen, Ami Paz, Joel Rybicki, and Stefan Schmid. Input-dynamic distributed algorithms for communication networks. In *Proc. ACM SIGMETRICS*, 2021.
- [170] Otavio Augusto de Oliveira Souza, Olga Goussevskaia, and Stefan Schmid. Cbnet: Minimizing adjustments in concurrent demand-aware tree networks. In *Proc. 35th IEEE International Parallel and Distributed Processing Symposium (IPDPS)*, 2021.
- [171] Julia Khamis, Stefan Schmid, and Ori Rottenstreich. Demand matrix optimization for offchain payments in blockchain. In *2021 IEEE International Conference on Blockchain and Cryptocurrency (ICBC)*, pages 1–9. IEEE, 2021.
- [172] Marco Chiesa, Roshan Sedar, Gianni Antichi, Michael Borokhovich, Andrzej Kamisinski, Georgios Nikolaidis, and Stefan Schmid. Fast reroute on programmable switches. In *IEEE/ACM Transactions on Networking (TON)*, 2021.
- [173] Maya Dotan, Yvonne-Anne Pignolet, Stefan Schmid, Saar Tochner, and Aviv Zohar. Survey on blockchain networking: Context, state-of-the-art, challenges. In *Proc. ACM Computing Surveys (CSUR)*, 2021.
- [174] Maciej Pacut, Mahmoud Parham, and Stefan Schmid. Optimal online balanced graph partitioning. In *Proc. IEEE INFOCOM*, 2021.
- [175] Apoorv Shukla, Kevin Hudemann, Zsolt Vagi, Lily Huegerich, Georgios Smaragdakis, Artur Hecker, Stefan Schmid, and Anja Feldmann. Fix with p6: Verifying programmable switches at runtime. In *Proc. IEEE INFOCOM*, 2021.
- [176] Klaus-Tycho Foerster, Andrzej Kamisinski, Yvonne-Anne Pignolet, Stefan Schmid, and Gilles Tredan. Grafting arborescences for extra resilience of fast rerouting schemes. In *Proc. IEEE INFOCOM*, 2021.
- [177] Monika Henzinger, Stefan Neumann, Harald Raecke, and Stefan Schmid. Tight bounds for online graph partitioning. In *Proc. ACM-SIAM Symposium on Discrete Algorithms (SODA)*, 2021.
- [178] Chen Avin and Stefan Schmid. Renets: Statically-optimal demand-aware networks. In *Proc. SIAM Symposium on Algorithmic Principles of Computer Systems (APOCS)*, 2021.
- [179] Klaus-Tycho Foerster, Juho Hirvonen, Yvonne-Anne Pignolet, Stefan Schmid, and Gilles Tredan. On the feasibility of perfect resilience with local fast failover. In *Proc. SIAM Symposium on Algorithmic Principles of Computer Systems (APOCS)*, 2021.
- [180] Tobias Forner, Harald Raecke, and Stefan Schmid. Online balanced repartitioning of dynamic communication patterns in polynomial time. In *Proc. SIAM Symposium on Algorithmic Principles of Computer Systems (APOCS)*, 2021.
- [181] Marco Chiesa, Andrzej Kamisinski, Jacek Rak, Gabor Retvari, and Stefan Schmid. A survey of fast-recovery mechanisms in packet-switched networks. *IEEE Communications Surveys and Tutorials (COMST)*, 2021.
- [182] Andrei Alexandru Agape, Madalin Claudiu Danceanu, Rene Rydhof Hansen, and Stefan Schmid. P4fuzz: Compiler fuzzer for dependable programmable dataplanes. In *Proc. 22nd International Conference on Distributed Computing and Networking (ICDCN)*, 2021.
- [183] Philipp Zabka, Klaus-Tycho Foerster, Stefan Schmid, and Christian Decker. Node classification and geographical analysis of the lightning cryptocurrency network. In *Proc. 22nd International Conference on Distributed Computing and Networking (ICDCN)*, 2021.

- [184] Mahmoud Parham, Klaus-Tycho Foerster, Petar Kosic, and Stefan Schmid. Maximally resilient replacement paths for a family of product graphs. In *Proc. OPODIS*, 2020.
- [185] Simon Scherrer, Markus Legner, Adrian Perrig, and Stefan Schmid. Incentivizing stable path selection in future internet architectures. In *Performance Evaluation (PEVA)*, Elsevier, 2020.
- [186] Elif Bozkaya, Klaus-Tycho Foerster, Stefan Schmid, and Berk Canberk. Airnet: Energy-aware deployment and scheduling of aerial networks. In *IEEE Transactions on Vehicular Technology (TVT)*, 2020.
- [187] Thomas Fenz, Klaus-Tycho Foerster, Stefan Schmid, and Anais Villedieu. Efficient non-segregated routing for reconfigurable demand-aware networks. In *Journal Elsevier Computer Communications (COMCOM)*, 2020.
- [188] Amaury Van Bemten, Nemanja Deric, Amir Varasteh, Stefan Schmid, Carmen Mas Machuca, Andreas Blenk, and Wolfgang Kellerer. Chameleon: Predictable latency and high utilization with queue-aware and adaptive source routing. In *Proc. 16th ACM International Conference on emerging Networking EXperiments and Technologies (CoNEXT)*, 2020.
- [189] Peter Gjol Jensen, Morten Konggaard, Dan Kristiansen, Stefan Schmid, Bernhard Clemens Schrenk, and Jiri Srba. Aalwines: A fast and quantitative what-if analysis tool for mpls networks. In *Proc. 16th ACM International Conference on emerging Networking EXperiments and Technologies (CoNEXT)*, 2020.
- [190] Saar Tochner and Stefan Schmid. On search friction of route discovery in offchain networks. In *Proc. IEEE International Conference on Blockchain (Blockchain)*, 2020.
- [191] Debasish Pattanayak, Klaus-Tycho Foerster, Partha Sarathi Mandal, and Stefan Schmid. Conic formation in presence of faulty robots. In *Proc. 16th International Symposium on Algorithms and Experiments for Wireless Sensor Networks (ALGOSENSORS)*, 2020.
- [192] Klaus-Tycho Foerster, Andrzej Kamisinski, Yvonne-Anne Pignolet, Stefan Schmid, and Gilles Tredan. Improved fast rerouting using postprocessing. In *IEEE Transactions on Dependable and Secure Computing (TDSC)*, 2020.
- [193] Saar Tochner, Aviv Zohar, and Stefan Schmid. Route hijacking and dos in off-chain networks. In *Proc. ACM Conference on Advances in Financial Technologies (AFT)*, 2020.
- [194] Klaus-Tycho Foerster, Juho Hirvonen, Yvonne-Anne Pignolet, Stefan Schmid, and Gilles Tredan. Brief announcement: What can(not) be perfectly rerouted locally. In *Proc. International Symposium on Distributed Computing (DISC)*, 2020.
- [195] Marcin Bienkowski, David Fuchssteiner, Jan Marcinkowski, and Stefan Schmid. Online dynamic b-matching with applications to reconfigurable datacenter networks. In *Proc. 38th International Symposium on Computer Performance, Modeling, Measurements and Evaluation (PERFORMANCE)*, 2020.
- [196] Mark Glavind, Niels Christensen, Jiri Srba, and Stefan Schmid. Latte: Improving the latency of transiently consistent network update schedules. In *Proc. 38th International Symposium on Computer Performance, Modeling, Measurements and Evaluation (PERFORMANCE)*, 2020.
- [197] Wenkai Dai, Klaus-Tycho Foerster, David Fuchssteiner, and Stefan Schmid. Load-optimization in reconfigurable networks: Algorithms and complexity of flow routing. In *Proc. 38th International Symposium on Computer Performance, Modeling, Measurements and Evaluation (PERFORMANCE)*, 2020.

- [198] Simon Scherrer, Markus Legner, Adrian Perrig, and Stefan Schmid. Incentivizing stable path selection in future internet architectures. In *Proc. 38th International Symposium on Computer Performance, Modeling, Measurements and Evaluation (PERFORMANCE)*, 2020.
- [199] Maciej Pacut, Mahmoud Parham, and Stefan Schmid. Brief announcement: Deterministic lower bound for dynamic balanced graph partitioning. In *Proc. ACM Symposium on Principles of Distributed Computing (PODC)*, 2020.
- [200] Debasish Pattanayak, Kaushik Mondal, Partha Sarathi Mandal, and Stefan Schmid. Area convergence of monoculus robots with additional capabilities. In *The Computer Journal (COMPJ)*, 2020.
- [201] Maya Dotan, Yvonne-Anne Pignolet, Saar Tochner, Stefan Schmid, and Aviv Zohar. Cryptocurrency networking: Context, state-of-the-art, challenges. In *Proc. 15th International Conference on Availability, Reliability and Security (ARES)*, 2020.
- [202] Chen Avin, Kaushik Mondal, and Stefan Schmid. Demand-aware network designs of bounded degree. In *Distributed Computing (DIST)*, 2020.
- [203] Chen Avin, Manya Ghobadi, Chen Griner, and Stefan Schmid. On the complexity of traffic traces and implications. In *Proc. of the ACM on Measurement and Analysis of Computing Systems (POMACS)*, 2020.
- [204] Balazs Nemeth, Yvonne-Anne Pignolet, Matthias Rost, Stefan Schmid, and Balazs Vass. Implications of routing coherence and consistency on network optimization. In *Proc. IFIP Networking*, 2020.
- [205] Yvonne-Anne Pignolet, Stefan Schmid, and Gilles Tredan. Implications of routing coherence and consistency on network optimization. In *Proc. IFIP Networking*, 2020.
- [206] Long Luo, Hongfang Yu, Klaus-Tycho Foerster, Max Noormohammadpour, and Stefan Schmid. Inter-datacenter bulk transfers: Trends and challenges. In *IEEE Network Magazine*, 2020.
- [207] Simon Scherrer, Adrian Perrig, and Stefan Schmid. The value of information in selfish routing. In *Proc. 27th International Colloquium on Structural Information and Communication Complexity (SIROCCO)*, 2020.
- [208] Samvid Dharanikota, Michael Toft Jensen, Sebastian Rom Kristensen, Mathias Sass Michno, Yvonne-Anne Pignolet, Rene Rydhof Hansen, and Stefan Schmid. Breeding unicorns: Developing trustworthy and scalable randomness beacons. In *PLOS ONE*, 2020.
- [209] Saeed Akhoondian Amiri, Klaus-Tycho Foerster, and Stefan Schmid. Walking through waypoints. In *Algorithmica (ALGO)*, 2020.
- [210] Matthias Rost and Stefan Schmid. On the hardness and inapproximability of virtual network embeddings. In *IEEE/ACM Transactions on Networking (TON)*, 2020.
- [211] Long Luo, Klaus-Tycho Foerster, Stefan Schmid, and Hongfang Yu. Deadline-aware multicast transfers in software-defined optical wide-area networks. In *IEEE Journal on Selected Areas in Communications (JSAC)*, 2020.
- [212] Apoorv Shukla, Seifeddine Fathalli, Thomas Zinner, Artur Hecker, and Stefan Schmid. P4consist: Towards consistent p4 sdns. In *IEEE Journal on Selected Areas in Communications (JSAC)*, 2020.
- [213] Chen Avin, Kaushik Mondal, and Stefan Schmid. Dynamically optimal self-adjusting single-source tree networks. In *Proc. Latin American Theoretical Informatics Symposium (LATIN)*, 2020.
- [214] Michael Feldmann, Christian Scheideler, and Stefan Schmid. Survey on algorithms for self-stabilizing overlay networks. In *ACM Computing Surveys (CSUR)*, 2020.

- [215] Chen Avin, Manya Ghobadi, Chen Griner, and Stefan Schmid. On the complexity of traffic traces and implications. In *Proc. ACM SIGMETRICS*, 2020.
- [216] Apoorv Shukla, Said Jawad Saidi, Stefan Schmid, Marco Canini, Thomas Zinner, and Anja Feldmann. Towards consistent sdn: A case for network state fuzzing. In *IEEE Transactions on Network and Service Management (TNSM)*, 2020.
- [217] Long Luo, Klaus-Tycho Foerster, Stefan Schmid, and Hongfang Yu. Splitcast: Optimizing multicast flows in reconfigurable datacenter networks. In *Proc. IEEE INFOCOM*, 2020.
- [218] Chen Avin, Iosif Salem, and Stefan Schmid. Working set theorems for routing in self-adjusting skip list networks. In *Proc. IEEE INFOCOM*, 2020.
- [219] Stefan Schmid Utz Nisslmueller, Klaus-Tycho Foerster and Christian Decker. Toward active and passive confidentiality attacks on cryptocurrency off-chain networks. In *Proc. 6th International Conference on Information Systems Security and Privacy (ICISSP)*, 2020.
- [220] Laura Schmid, Krishnendu Chatterjee, and Stefan Schmid. The evolutionary price of anarchy: Locally bounded agents in a dynamic virus game. In *Proc. 22nd International Conference on Principles of Distributed Systems (OPODIS)*, 2019.
- [221] Chen Avin, Iosif Salem, and Stefan Schmid. Brief announcement: On self-adjusting skip list networks. In *Proc. 33rd International Symposium on Distributed Computing (DISC)*, 2019.
- [222] Levente Csikor, Dinil Mon Divakaran, Min Suk Kang, Attila Korosi, Balazs Sonkoly, David Haja, Dimitrios Pezaros, Stefan Schmid, and Gabor Retvari. Tuple space explosion: A denial-of-service attack against a software packet classifier. In *Proc. ACM International Conference on emerging Networking EXperiments and Technologies (CoNEXT)*, 2019.
- [223] Marco Chiesa, Roshan Sedar, Gianni Antichi, Michael Borokhovich, Andrzej Kamisinski, Georgios Nikolaidis, and Stefan Schmid. Purr: A primitive for reconfigurable fast reroute. In *Proc. ACM International Conference on emerging Networking EXperiments and Technologies (CoNEXT)*, 2019.
- [224] Amaury Van Bemten, Nemanja Deric, Johannes Zerwas, Andreas Blenk, Stefan Schmid, and Wolfgang Kellerer. Loko: Predictable latency in small networks. In *Proc. ACM International Conference on emerging Networking EXperiments and Technologies (CoNEXT)*, 2019.
- [225] Monika Henzinger, Stefan Neumann, and Stefan Schmid. Efficient distributed workload (re-)embedding. In *Proc. of the ACM on Measurement and Analysis of Computing Systems (POMACS)*, Vol. 3, No. 1, Article 13, 2019.
- [226] Chen Avin, Ingo van Duijn, and Stefan Schmid. Self-adjusting linear networks. In *Proc. 21st International Symposium on Stabilization, Safety, and Security of Distributed Systems (SSS)*, 2019.
- [227] Matthias Rost and Stefan Schmid. Virtual network embedding approximations: Leveraging randomized rounding. In *Proc. IEEE/ACM Transactions on Networking (ToN)*, 2019.
- [228] Or Raz, Chen Avin, and Stefan Schmid. Nap: Network-aware data partitions for efficient distributed processing. In *Proc. IEEE International Symposium on Network Computing and Applications (NCA)*, 2019.
- [229] Klaus-Tycho Foerster and Stefan Schmid. Distributed consistent network updates in sdn: Local verification for global guarantees. In *Proc. IEEE International Symposium on Network Computing and Applications (NCA)*, 2019.
- [230] Amaury Van Bemten, Nemanja Deric, Amir Varasteh, Andreas Blenk, Stefan Schmid, and Wolfgang Kellerer. Empirical predictability study of sdn switches. In *Proc. ACM/IEEE Symposium on Architectures for Networking and Communications Systems (ANCS)*, 2019.

- [231] Kashyap Thimmaraju, Liron Schiff, and Stefan Schmid. Preacher: Network policy checker for adversarial environments. In *Proc. 38th International Symposium on Reliable Distributed Systems (SRDS)*, 2019.
- [232] Klaus-Tycho Foerster, Andrzej Kamisinski, Yvonne-Anne Pignolet, Stefan Schmid, and Gilles Tredan. Improved fast rerouting using postprocessing. In *Proc. 38th International Symposium on Reliable Distributed Systems (SRDS)*, 2019.
- [233] Harald Raecke and Stefan Schmid. Compact oblivious routing. In *Proc. European Symposium on Algorithms (ESA)*, 2019.
- [234] Johannes Zerwas, Patrick Kalmbach, Stefan Schmid, and Andreas Blenk. Ismael: Using machine learning to predict acceptance of virtual clusters in data centers. In *IEEE Transactions on Network and Service Management (Special Issue)*, 2019.
- [235] Mu He, Andreas Blenk, Stefan Schmid, and Wolfgang Kellerer. Toward consistent state management of adaptive programmable networks based on p4. In *Proc. ACM SIGCOMM Workshop on Networking for Emerging Applications and Technologies (NEAT)*, 2019.
- [236] Klaus-Tycho Foerster, Janne Korhonen, Joel Rybicki, and Stefan Schmid. Brief announcement: Does preprocessing help under congestion? In *Proc. ACM Symposium on Principles of Distributed Computing (PODC)*, 2019.
- [237] Samvid Dharanikota, Rene Rydhof Hansen, Michael Jensen, Sebastian Ro Kristensen, Matthias Sass Michno, Yvonne-Anne Pignolet, and Stefan Schmi. Breeding unicorns: Developing trustworthy and scalable randomness beacons. In *Proc. IEEE International Conference on Blockchain (Blockchain)*, 2019.
- [238] Johannes Zerwas, Patrick Kalmbach, Laurenz Henkel, Gabor Retvari, Wolfgang Kellerer, Andreas Blenk, and Stefan Schmid. Netboa: Self-driving network benchmarking. In *Proc. ACM SIGCOMM Workshop on Network Meets AI and ML (NetAI)*, 2019.
- [239] Apoorv Shukla, Kevin Nico Hudemann, Artur Hecker, and Stefan Schmid. Runtime verification of p4 switches with reinforcement learning. In *Proc. ACM SIGCOMM Workshop on Network Meets AI and ML (NetAI)*, 2019.
- [240] Kashyap Thimmaraju, Saad Hermak, Gabor Retvari, and Stefan Schmid. mswitch: Bringing multi-tenancy to virtual switches. In *Proc. USENIX Annual Technical Conference (ATC)*, 2019.
- [241] Klaus-Tycho Foerster and Stefan Schmid. Survey of reconfigurable data center networks: Enablers, algorithms, complexity. In *SIGACT News*, 2019.
- [242] Klaus-Tycho Foerster, Maciej Pacut, and Stefan Schmid. On the complexity of non-segregated routing in reconfigurable data center architectures. In *ACM SIGCOMM Computer Communication Review (CCR)*, 2019.
- [243] Saeed Amiri, Stefan Schmid, and Sebastian Siebertz. Distributed dominating set approximations beyond planar graphs. In *ACM Transactions on Algorithms (TALG)*, 2019.
- [244] Chen Avin, Marcin Bienkowski, Andreas Loukas, Maciej Pacut, and Stefan Schmid. Dynamic balanced graph partitioning. In *SIAM J. Discrete Math (SIDMA)*, 2019.
- [245] Wolfgang Kellerer, Patrick Kalmbach, Andreas Blenk, Arsany Basta, Martin Reisslein, and Stefan Schmid. Adaptable and data-driven softwarized networks: Review, opportunities, and challenges. In *Proceedings of the IEEE (PIEEE)*, 2019.

- [246] Klaus-Tycho Foerster, Andrzej Kamisinski, Yvonne-Anne Pignolet, Stefan Schmid, and Gilles Tredan. Bonsai: Efficient fast failover routing using small arborescences. In *Proc. 49th IEEE/IFIP International Conference on Dependable Systems and Networks (DSN)*, 2019.
- [247] Gregor Bankhamer, Robert Elsaesser, and Stefan Schmid. Local fast rerouting with low congestion: A randomized approach. In *Proc. 27th IEEE International Conference on Network Protocols (ICNP)*, 2020.
- [248] Thomas Fenz, Klaus-Tycho Foerster, Stefan Schmid, and Anais Villedieu. Efficient non-segregated routing for reconfigurable demand-aware networks. In *Proc. IFIP Networking*, 2019.
- [249] Saeed Amiri, Szymon Dudycz, Mahmoud Parham, Stefan Schmid, and Sebastian Wiederrech. On polynomial-time congestion-free software-defined network updates. In *Proc. IFIP Networking*, 2019.
- [250] Andreas Blenk, Arsany Basta, Wolfgang Kellerer, and Stefan Schmid. On the impact of the network hypervisor on virtual network performance. In *Proc. IFIP Networking*, 2019.
- [251] Fabien Geyer and Stefan Schmid. Deepmpls: Fast analysis of mpls configurations using deep learning. In *Proc. IFIP Networking*, 2019.
- [252] Matthias Rost, Elias Doehne, and Stefan Schmid. Parametrized complexity of virtual network embeddings: Dynamic & linear programming approximations. In *ACM SIGCOMM Computer Communication Review (CCR)*, 2018.
- [253] Monika Henzinger, Stefan Neumann, and Stefan Schmid. Efficient distributed workload (re-)embedding. In *Proc. ACM SIGMETRICS / IFIP PERFORMANCE*, 2019.
- [254] Chen Avin, Kaushik Mondal, and Stefan Schmid. Demand-aware network design with minimal congestion and route lengths. In *Proc. IEE INFOCOM*, 2019.
- [255] Bruna Peres, Otavio Souza, Olga Goussevskaia, Stefan Schmid, and Chen Avin. Distributed self-adjusting tree networks. In *Proc. IEE INFOCOM*, 2019.
- [256] Mengxue Liu, Andrea Richa, Stefan Schmid, and Matthias Rost. A constant approximation for maximum throughput multicommodity routing. In *Proc. IEE INFOCOM*, 2019.
- [257] Klaus-Tycho Foerster, Juho Hirvonen, Stefan Schmid, and Jukka Suomela. On the power of preprocessing in decentralized network optimization. In *Proc. IEE INFOCOM*, 2019.
- [258] Klaus-Tycho Foerster, Yvonne-Anne Pignolet, Stefan Schmid, and Gilles Tredan. Casa: Congestion and stretch aware static fast rerouting. In *Proc. IEE INFOCOM*, 2019.
- [259] Klaus-Tycho Foerster, Mahmoud Parham, Stefan Schmid, and Tao Wen. Local fast segment rerouting on hypercubes. In *22nd International Conference on Principles of Distributed Systems (OPODIS)*, 2018.
- [260] Klaus-Tycho Foerster, Stefan Schmid, and Stefano Vissicchio. Survey of consistent software-defined network updates. In *IEEE Communications Surveys and Tutorials (COMST)*, 2018.
- [261] Oliver Hohlfeld, James Kempf, Martin Reisslein, Stefan Schmid, and Nadir Shah. Scalability issues and solutions for software defined networks (editorial). In *IEEE JSAC Special Issue on Scalability Issues and Solutions for Software Defined Networks (JSAC)*, 2019.
- [262] Chen Avin and Stefan Schmid. Toward demand-aware networking: A theory for self-adjusting networks. In *ACM SIGCOMM Computer Communication Review (CCR)*, 2018.

- [263] Jesper Stenbjerg Jensen, Troels Beck Krogh, Jonas Sand Madsen, Stefan Schmid, Jiri Srba, and Marc Tom Thorgersen. P-rex: Fast verification of mpls networks with multiple link failures. In *14th ACM International Conference on emerging Networking EXperiments and Technologies (CoNEXT)*, 2018.
- [264] Levente Csikor, Christian Rothenberg, Dimitrios P. Pezaros, Stefan Schmid, Laszlo Toka, and Gabor Retvari. Policy injection: A cloud dataplane dos attack. In *ACM SIGCOMM 2018 Demo Paper*, 2018.
- [265] Andreas Blenk, Arsany Basta, Laurenz Henkel, Johannes Zerwas, Wolfgang Kellerer, and Stefan Schmid. Perfbench: A tool for predictability analysis in multi-tenant software-defined networks. In *ACM SIGCOMM 2018 Poster Paper*, 2018.
- [266] Patrick Kalmbach, Lion Gleiter, Johannes Zerwas, Andreas Blenk, Wolfgang Kellerer, and Stefan Schmid. Modeling ip-to-ip communication using the weighted stochastic block model. In *ACM SIGCOMM 2018 Poster Paper*, 2018.
- [267] Tamas Lukovszki, Matthias Rost, and Stefan Schmid. Approximate and incremental network function placement. In *Journal of Parallel and Distributed Computing (JPDC)*, 2018.
- [268] Chen Avin, Louis Cohen, Mahmoud Parham, and Stefan Schmid. Competitive clustering of stochastic communication patterns on a ring. In *Journal of Computing*, 2018.
- [269] Klaus-Tycho Foerster, Monia Ghobadi, and Stefan Schmid. Characterizing the algorithmic complexity of reconfigurable data center architectures. In *Proc. ACM/IEEE Symposium on Architectures for Networking and Communications Systems (ANCS)*, 2018.
- [270] Liron Schiff, Ofri Ziv, Manfred Jaeger, and Stefan Schmid. Netslicer: Automated and traffic-pattern based application clustering in datacenters. In *Proc. ACM SIGCOMM 2018 Workshop on Big Data Analytics and Machine Learning for Data Communication Networks (Big-DAMA)*, 2018.
- [271] Kashyap Thimmaraju, Gabor Retvari, and Stefan Schmid. Virtual network isolation: Are we there yet? In *Proc. ACM SIGCOMM 2018 Workshop on Security in Softwarized Networks: Prospects and Challenges (SecSon)*, 2018.
- [272] Roshan Sedar, Michael Borokhovich, Marco Chiesa, Gianni Antichi, and Stefan Schmid. Supporting emerging applications with low-latency failover in p4. In *Proc. ACM SIGCOMM 2018 Workshop on Networking for Emerging Applications and Technologies (NEAT)*, 2018.
- [273] Patrick Kalmbach, Johannes Zerwas, Peter Babarczi, Andreas Blenk, Wolfgang Kellerer, and Stefan Schmid. Empowering self-driving networks. In *Proc. ACM SIGCOMM 2018 Workshop on Self-Driving Networks (SDN)*, 2018.
- [274] Saeed Akhoondian Amiri, Szymon Dudycz, Stefan Schmid, and Sebastian Wiederrecht. Congestion-free rerouting of flows on dags. In *Proc. 45th International Colloquium on Automata, Languages, and Programming (ICALP)*, 2018.
- [275] Jiaqi Zheng, Bo Li, Chen Tian, Klaus-Tycho Foerster, Stefan Schmid, Guihai Chen, and Jie Wu. Scheduling congestion-free updates of multiple flows with chronicle in timed sdns. In *Proc. IEEE ICDCS*, 2018.
- [276] Marco Canini, Iosif Salem, Liron Schiff, Elad Michael Schiller, and Stefan Schmid. Renaissance: A self-stabilizing distributed sdn control plane. In *Proc. IEEE ICDCS*, 2018.
- [277] Saeed Akhoondian Amiri, Klaus-Tycho Foerster, Riko Jacob, Mahmoud Parham, and Stefan Schmid. Waypoint routing in special networks. In *Proc. IFIP Networking*, 2018.

- [278] Robert Kroesche, Kashyap Thimmaraju, Liron Schiff, and Stefan Schmid. I dpid it my way! a covert timing channel in software-defined networks. In *Proc. IFIP Networking*, 2018.
- [279] Matthias Rost and Stefan Schmid. Charting the complexity landscape of virtual network embeddings. In *Proc. IFIP Networking*, 2018.
- [280] Johannes Zerwas, Patrick Kalmbach, Carlo Fuerst, Arne Ludwig, Andreas Blenk, Wolfgang Kellerer, and Stefan Schmid. Ahab: Data-driven virtual cluster hunting. In *Proc. IFIP Networking*, 2018.
- [281] Matthias Rost and Stefan Schmid. Virtual network embedding approximations: Leveraging randomized rounding. In *Proc. IFIP Networking*, 2018.
- [282] Balazs Nemeth, Mark Szalay, Janos Doka, Matthias Rost, Stefan Schmid, Laszlo Toka, and Balazs Sonkoly. Fast and efficient network service embedding method with adaptive offloading to the edge. In *Proc. INFOCOM Workshop on Integrating Edge Computing, Caching, and Offloading in Next Generation Networks*, 2018.
- [283] Kashyap Thimmaraju, Bhargava Shastry, Tobias Fiebig, Felicitas Hetzelt, Jean-Pierre Seifert, Anja Feldmann, and Stefan Schmid. Taking control of sdn-based cloud systems via the data plane. In *Proc. ACM Symposium on SDN Research (SOSR)*, 2018.
- [284] Klaus-Tycho Foerster, Yvonne-Anne Pignolet, Stefan Schmid, and Gilles Tredan. Local fast failover routing with low stretch. In *Proc. ACM SIGCOMM Computer Communication Review (CCR)*, 2018.
- [285] Saeed Akhoondian Amiri, Klaus-Tycho Foerster, Riko Jacob, and Stefan Schmid. Charting the algorithmic complexity of waypoint routing. In *Proc. ACM SIGCOMM Computer Communication Review (CCR)*, 2018.
- [286] Klaus-Tycho Foerster, Mahmoud Parham, Marco Chiesa, and Stefan Schmid. Ti-mfa: Keep calm and reroute segments fast. In *Proc. IEEE Global Internet Symposium (GI)*, 2018.
- [287] Michael Borokhovich, Clement Rault, Liron Schiff, and Stefan Schmid. The show must go on: Fundamental data plane connectivity services for dependable sdns. In *Elsevier Computer Communications (COMCOM)*, 2018.
- [288] Saeed Akhoondian Amiri, Klaus-Tycho Foerster, and Stefan Schmid. Walking through waypoints. In *Proc. 13th Latin American Theoretical Informatics Symposium (LATIN)*, 2018.
- [289] Patrick Kalmbach, Andreas Blenk, Wolfgang Kellerer, and Stefan Schmid. Themis: A data-driven approach to bot detection (short abstract). In *Proc. IEE INFOCOM*, 2018.
- [290] Arne Ludwig, Szymon Dudycz, Matthias Rost, and Stefan Schmid. Transiently policy-compliant network updates. In *Proc. IEEE/ACM Transactions on Networking (ToN)*, 2018.
- [291] Michael Borokhovich, Yvonne-Anne Pignolet, Gilles Tredan, and Stefan Schmid. Load-optimal local fast rerouting for dense networks. In *Proc. IEEE/ACM Transactions on Networking (ToN)*, 2018.
- [292] Arsany Basta, Andreas Blenk, Szymon Dudycz, Arne Ludwig, and Stefan Schmid. Efficient loop-free rerouting of multiple sdn flows. In *Proc. IEEE/ACM Transactions on Networking (ToN)*, 2018.
- [293] Marcin Bienkowski, Nadi Sarrar, Stefan Schmid, and Steve Uhlig. Online aggregation of the forwarding information base: Accounting for locality and churn. In *Proc. IEEE/ACM Transactions on Networking (ToN)*, 2018.
- [294] Carlo Fuerst, Stefan Schmid, Lalith Suresh, and Paolo Costa. Kraken: Online and elastic resource reservations for cloud datacenters. In *Proc. IEEE/ACM Transactions on Networking (ToN)*, 2018.

- [295] Klaus-Tycho Foerster, Arne Ludwig, Jan Marcinkowski, and Stefan Schmid. Loop-free route updates for software-defined networks. In *Proc. IEEE/ACM Transactions on Networking (ToN)*, 2018.
- [296] Yvonne Anne Pignolet, Stefan Schmid, and Gilles Tredan. Tomographic node placement strategies and the impact of the routing model. In *Proc. ACM SIGMETRICS*, 2018.
- [297] Juho Hirvonen, Joel Rybicki, Stefan Schmid, and Jukka Suomela. Large cuts with local algorithms on triangle-free graphs. In *The Electronic Journal of Combinatorics (EJC)*, 2017.
- [298] Jiaqi Zheng, Guihai Chen, Stefan Schmid, Haipeng Dai, and Jie Wu. Consistent network updates in timed sdns. In *IEEE JSAC Special Issue on Emerging Technologies in Software-Driven Communication*, 2017.
- [299] Andreas Blenk, Patrick Kalmbach, Johannes Zerwas, Michael Jarschel, Stefan Schmid, and Wolfgang Kellerer. Neurovine: A neural preprocessor for your virtual network embedding algorithm. In *Proc. IEE INFOCOM*, 2018.
- [300] Stefan Schmid and Jiri Srba. Polynomial-time what-if analysis for prefix-manipulating mpls networks. In *Proc. IEE INFOCOM*, 2018.
- [301] Yvonne Anne Pignolet, Stefan Schmid, and Gilles Tredan. Tomographic node placement strategies and the impact of the routing model. In *Proc. of the ACM on Measurement and Analysis of Computing Systems (POMACS)*, Vol. 1, No. 2, Article 42, 2017.
- [302] Kashyap Thimmaraju, Bhargava Shastry, Tobias Fiebig, Felicitas Hetzelt, Jean-Pierre Seifert, Anja Feldmann, and Stefan Schmid. The vamp attack: Taking control of cloud systems via the unified packet parser. In *Proc. 9th ACM Cloud Computing Security Workshop (CCSW)*, 2017.
- [303] Mu He, Patrick Kalmbach, Andreas Blenk, Stefan Schmid, and Wolfgang Kellerer. Algorithm-data driven optimization of adaptive communication networks. In *Proc. IEEE ICNP Workshop on Machine Learning and Artificial Intelligence in Computer Networks*, 2017.
- [304] Julius Schulz-Zander, Carlos Mayer, Bogdan Ciobotaru, Raphael Lisicki, Stefan Schmid, and Anja Feldmann. Unified programmability of virtualized network functions and software-defined wireless networks. In *IEEE Transactions on Network and Service Management (TNSM)*, 2017.
- [305] Adrian Ogierman, Andrea Richa, Christian Scheideler, Stefan Schmid, and Jin Zhang. Sade: Competitive mac under adversarial sinr. In *Journal Distributed Computing (DIST)*, 2017.
- [306] Klaus-Tycho Foerster, Mahmoud Parham, and Stefan Schmid. A walk in the clouds: Routing through vnfs on bidirected networks. In *Proc. 3rd International Workshop on Algorithmic Aspects of Cloud Computing (ALGO CLOUD)*, 2017.
- [307] Chen Avin, Kaushik Mondal, and Stefan Schmid. Demand-aware network designs of bounded degree. In *Proc. International Symposium on Distributed Computing (DISC)*, 2017.
- [308] Bruna Peres, Olga Goussevskaia, Stefan Schmid, and Chen Avin. Brief announcement: Distributed splaynets. In *Proc. International Symposium on Distributed Computing (DISC)*, 2017.
- [309] Carlo Fuerst, Maciej Pacut, and Stefan Schmid. Data locality and replica aware virtual cluster embeddings. In *Journal Theoretical Computer Science (TCS)*, Elsevier, 2017.
- [310] Andreas Blenk, Patrick Kalmbach, Stefan Schmid, and Wolfgang Kellerer. o’zapft is: Tap your network algorithm’s big data! In *Proc. ACM SIGCOMM 2017 Workshop on Big Data Analytics and Machine Learning for Data Communication Networks (Big-DAMA)*, 2017.

- [311] Marcel Bloecher, Malte Viering, Stefan Schmid, and Patrick Eugster. The grand cru challenge. In *Proc. ACM SIGCOMM 2017 International Workshop on Hot Topics in Container Networking and Networked Systems (HotConNet)*, 2017.
- [312] Marcin Bienkowski, Jan Marcinkowski, Maciej Pacut, Stefan Schmid, and Aleksandra Spyra. Online tree caching. In *Proc. 29th ACM Symposium on Parallelism in Algorithms and Architectures (SPAA)*, 2017.
- [313] Yvonne-Anne Pignolet, Stefan Schmid, and Gilles Tredan. Load-optimal local fast rerouting for resilient networks. In *Proc. 47th IEEE/IFIP International Conference on Dependable Systems and Networks (DSN)*, 2017.
- [314] Jiaqi Zheng, Guihai Chen, Stefan Schmid, Haipeng Dai, and Jie Wu. Scheduling congestion- and loop-free network update in timed sdns. In *Proc. 37th IEEE International Conference on Distributed Computing Systems (ICDCS)*, 2017.
- [315] Chen Avin, Louis Cohen, and Stefan Schmid. Competitive clustering of stochastic communication patterns on the ring. In *Proc. 5th International Conference on Networked Systems (NETYS)*, 2017.
- [316] Debasish Pattanayak, Kaushik Mondal, Partha Sarathi Mandal, and Stefan Schmid. Convergence of even simpler robots without location information. In *Proc. 5th International Conference on Networked Systems (NETYS)*, 2017.
- [317] Kashyap Thimmaraju, Liron Schiff, and Stefan Schmid. Outsmarting network security with sdn teleportation. In *Proc. 2nd IEEE European Symposium on Security and Privacy (EuroSP)*, 2017.
- [318] Matthias Hollick, Cristina Nita-Rotaru, Panagiotis Papadimitratos, Adrian Perrig, and Stefan Schmid. Toward a taxonomy and attacker model for secure routing protocols. In *Proc. ACM SIGCOMM Computer Communication Review (CCR)*, 2017.
- [319] Kim G. Larsen, Stefan Schmid, and Bingtian Xue. Wnetkat: A weighted sdn programming and verification language. In *Proc. 20th International Conference on Principles of Distributed Systems (OPODIS)*, 2016.
- [320] Julius Schulz-Zander, Raphael Lisicki, Stefan Schmid, and Anja Feldmann. Secuspot: Toward cloud-assisted secure multi-tenant wifi hotspot infrastructures. In *Proc. ACM CoNEXT Workshop on Cloud-Assisted Networking (CAN)*, 2016.
- [321] Balazs Nemeth, Balazs Sonkoly, Matthias Rost, and Stefan Schmid. Efficient service graph embedding: A practical approach. In *Proc. IEEE NFV-SDN Workshop O4SDI*, 2016.
- [322] Kim G. Larsen, Stefan Schmid, and Bingtian Xue. Wnetkat: Programming and verifying weighted software-defined networks. In *Proc. 28th Nordic Workshop on Programming Theory (NWPT)*, 2016.
- [323] Julius Schulz-Zander, Stefan Schmid, James Kempf, Roberto Riggio, and Anja Feldmann. Legofi the wifi building blocks! the case for a modular wifi architecture. In *Proc. ACM MOBICOM Workshop on Mobility in the Evolving Internet Architecture (MobiArch)*, 2016.
- [324] Chen Avin, Andreas Loukas, Maciej Pacut, and Stefan Schmid. Online balanced repartitioning. In *Proc. 30th International Symposium on Distributed Computing (DISC)*, 2016.
- [325] Saeed Akhoondian Amiri and Stefan Schmid. Brief announcement: A log-star-time local mds approximation scheme for bounded genus graphs. In *Proc. 30th International Symposium on Distributed Computing (DISC)*, 2016.
- [326] Christina Rickmann, Christoph Wagner, Uwe Nestmann, and Stefan Schmid. Topological self-stabilization with name-passing process calculi. In *Proc. 27th International Conference on Concurrency Theory (CONCUR)*, 2016.

- [327] Liron Schiff, Kashyap Thimmaraju, and Stefan Schmid. Routing-verification-as-a-service (rvaas): Trustworthy routing despite insecure providers. In *Proc. IEEE/IFIP DSN Workshop on Dependability Issues on SDN and NFV (DISN)*, 2016.
- [328] Liron Schiff, Stefan Schmid, and Marco Canini. Ground control to major faults: Towards a fault tolerant and adaptive sdn control network. In *Proc. IEEE/IFIP DSN Workshop on Dependability Issues on SDN and NFV (DISN)*, 2016.
- [329] Anja Feldmann, Philipp Heyder, Michael Kreutzer, Stefan Schmid, Jean-Pierre Seifert, Haya Shulman, Kashyap Thimmaraju, Michael Waidner, and Jens Sieberg. Netco: Reliable routing with unreliable routers. In *Proc. IEEE/IFIP DSN Workshop on Dependability Issues on SDN and NFV (DISN)*, 2016.
- [330] Saeed Amiri, Stefan Schmid, and Sebastian Siebertz. A local constant factor mds approximation for bounded genus graphs. In *Proc. ACM Symposium on Principles of Distributed Computing (PODC)*, 2016.
- [331] Guy Even, Matthias Rost, and Stefan Schmid. An approximation algorithm for path computation and function placement in sdns. In *Proc. 23rd International Colloquium on Structural Information and Communication Complexity (SIROCCO)*, 2016.
- [332] Saeed Amiri, Arne Ludwig, Jan Marcinkowski, and Stefan Schmid. Transiently consistent sdn updates: Being greedy is hard. In *Proc. 23rd International Colloquium on Structural Information and Communication Complexity (SIROCCO)*, 2016.
- [333] Szymon Dudycz, Arne Ludwig, and Stefan Schmid. Can’t touch this: Consistent network updates for multiple policies. In *Proc. 46th IEEE/IFIP International Conference on Dependable Systems and Networks (DSN)*, 2016.
- [334] Arne Ludwig, Szymon Dudycz, Matthias Rost, and Stefan Schmid. Transiently secure network updates. In *Proc. ACM SIGMETRICS*, 2016.
- [335] Liron Schiff and Stefan Schmid. Study the past if you would define the future: Implementing secure multi-party sdn updates. In *Proc. IEEE SwSTE*, 2016.
- [336] Liron Schiff and Stefan Schmid. Pri: Privacy preserving inspection of encrypted network traffic. In *Proc. IEEE Security&Privacy Workshop on Research for Insider Threats (WRIT)*, 2016.
- [337] Carlo Fuerst, Stefan Schmid, Lalith Suresh, and Paolo Costa. Kraken: Online and elastic resource reservations for multi-tenant datacenters. In *Proc. 35th IEEE Conference on Computer Communications (INFOCOM)*, 2016.
- [338] Vasileios Kotronis, Rowan Kloti, Matthias Rost, Panagiotis Georgopoulos, Bernhard Ager, Stefan Schmid, and Xenofontas Dimitropoulos. Stitching inter-domain paths over ixps. In *Proc. ACM Symposium on SDN Research (SOSR)*, 2016.
- [339] Tamas Lukovszki, Matthias Rost, and Stefan Schmid. It’s a match! near-optimal and incremental middlebox deployment. *ACM SIGCOMM Computer Communication Review (CCR)*, 2016.
- [340] Yvonne Anne Pignolet, Stefan Schmid, and Gilles Tredan. Upper and lower bounds for deterministic broadcast in powerline communication networks. *Springer Distributed Computing (DIST)*, 2016.
- [341] Liron Schiff, Petr Kuznetsov, and Stefan Schmid. In-band synchronization for distributed sdn control planes. *ACM SIGCOMM Computer Communication Review (CCR)*, 2016.
- [342] Sahel Sakhaf, Wouter Tavernier, Matthias Rost, Stefan Schmid, Didier Colle, Mario Pickavet, and Piet Demeester. Network service chaining with optimized network function embedding supporting service decompositions. In *Journal Computer Networks (COMNET)*, Elsevier, 2016.

- [343] Michael Markovitch and Stefan Schmid. Shear: A highly available and flexible network architecture: Marrying distributed and logically centralized control planes. In *Proc. 23rd IEEE International Conference on Network Protocols (ICNP)*, 2015.
- [344] Carlo Fuerst, Maciej Pacut, Paolo Costa, and Stefan Schmid. How hard can it be? understanding the complexity of replica aware virtual cluster embeddings. In *Proc. 23rd IEEE International Conference on Network Protocols (ICNP)*, 2015.
- [345] Juho Hirvonen, Joel Rybicki, Stefan Schmid, and Jukka Suomela. Large cuts with local algorithms on triangle-free graphs. In *Proc. 29th International Symposium on Distributed Computing (DISC)*, 2015.
- [346] Julius Schulz-Zander, Carlos Mayer, Bogdan Ciobotaru, Stefan Schmid, Anja Feldmann, and Roberto Riggio. Programming the home and enterprise wifi with opensdwn (demo). In *Proc. ACM SIGCOMM*, 2015.
- [347] Liron Schiff, Stefan Schmid, and Marco Canini. Medieval: Towards a self-stabilizing, plug & play, in-band sdn control network. In *Proc. ACM Sigcomm Symposium on SDN Research (SOSR)*, 2015.
- [348] Arne Ludwig, Jan Marcinkowski, and Stefan Schmid. Scheduling loop-free network updates: It’s good to relax! In *Proc. ACM Symposium on Principles of Distributed Computing (PODC)*, 2015.
- [349] Tamas Lukovszki and Stefan Schmid. Online admission control and embedding of service chains. In *Proc. 22nd International Colloquium on Structural Information and Communication Complexity (SIROCCO)*, 2015.
- [350] Matthias Rost, Carlo Fuerst, and Stefan Schmid. Beyond the stars: Revisiting virtual cluster embeddings. In *Proc. ACM SIGCOMM Computer Communication Review (CCR)*, 2015.
- [351] Julius Schulz-Zander, Carlos Mayer, Bogdan Ciobotaru, Stefan Schmid, and Anja Feldmann. Opensdwn: Programmatic control over home and enterprise wifi. In *Proc. ACM Sigcomm Symposium on SDN Research (SOSR)*, 2015.
- [352] Arne Ludwig and Stefan Schmid. Distributed cloud market: Who benefits from specification flexibilities? In *Proc. 3rd SIGMETRICS Workshop on Distributed Cloud Computing (DCC) and Special Issue ACM Performance Evaluation Review (PER)*, 2015.
- [353] Xinhui Hu, Arne Ludwig, Andrea Richa, and Stefan Schmid. Competitive strategies for online cloud resource allocation with discounts: The 2-dimensional parking permit problem. In *Proc. 35th IEEE International Conference on Distributed Computing Systems (ICDCS)*, 2015.
- [354] Carlo Fuerst, Stefan Schmid, Lalith Suresh, and Paolo Costa. Kraken: Towards elastic performance guarantees in multi-tenant data centers. In *Proc. ACM SIGMETRICS*, 2015.
- [355] Vasileios Kotronis, Rowan Kloeti, Matthias Rost, Panagiotis Georgopoulos, Bernhard Ager, Stefan Schmid, and Xenofontas Dimitropoulos. Investigating the potential of the inter-ixp multigraph for the provisioning of guaranteed end-to-end services. In *Proc. ACM SIGMETRICS*, 2015.
- [356] Stefan Schmid, Chen Avin, Christian Scheideler, Michael Borokhovich, Bernhard Haeupler, and Zvi Lotker. Splaynet: Towards locally self-adjusting networks. *IEEE/ACM Transactions on Networking (ToN)*, 2016.
- [357] Lalith Suresh, Marco Canini, Stefan Schmid, and Anja Feldmann. C3: Cutting tail latency in cloud data stores via adaptive replica selection. In *Proc. 12th USENIX Symposium on Networked Systems Design and Implementation (NSDI)*, 2015.
- [358] Marco Canini, Petr Kuznetsov, Dan Levin, and Stefan Schmid. A distributed and robust sdn control plane for transactional network updates. In *Proc. 34th IEEE Conference on Computer Communications (INFOCOM)*, 2015.

- [359] Yvonne Coady, Oliver Hohlfeld, James Kempf, Rick McGeer, and Stefan Schmid. Distributed cloud computing: Applications, status quo, and challenges (editorial). In *Proc. ACM SIGCOMM Computer Communication Review (CCR)*, 2015.
- [360] Arne Ludwig, Matthias Rost, Damien Foucard, and Stefan Schmid. Good network updates for bad packets: Waypoint enforcement beyond destination-based routing policies. In *Proc. ACM Workshop on Hot Topics in Networks (HotNets)*, 2014.
- [361] Liron Schiff, Michael Borokhovich, and Stefan Schmid. Reclaiming the brain: Useful openflow functions in the data plane. In *Proc. ACM Workshop on Hot Topics in Networks (HotNets)*, 2014.
- [362] Marcin Bienkowski, Leszek Gasieniec, Marek Klonowski, Mirosław Korzeniowski, Bernard Mans, Stefan Schmid, and Roger Wattenhofer. Distributed alarming in the on-duty and off-duty models. *IEEE/ACM Transactions on Networking (ToN)*, 2015.
- [363] Marco Canini, Anja Feldmann, Dan Levin, Fabian Schaffert, and Stefan Schmid. Panopticon: Incremental deployment of software-defined networking. *IEEE Computer*, 2014.
- [364] Riko Jacob, Andrea Richa, Christian Scheideler, Stefan Schmid, and Hanjo Taeubig. Skip+: A self-stabilizing skip graph. *Journal of the ACM (JACM)*, 2014.
- [365] Nadi Sarrar, Robert Wuttke, Stefan Schmid, Marcin Bienkowski, and Steve Uhlig. Leveraging locality for fib aggregation. In *Proc. IEEE GLOBECOM*, 2014.
- [366] Yvonne Anne Pignolet, Stefan Schmid, and Gilles Tredan. Colorcast: Deterministic broadcast in powerline networks with uncertainties. In *Proc. 5th IEEE International Conference on Smart Grid Communications (SmartGridComm)*, 2014.
- [367] P. Skoldstrom et al. Towards unified programmability of cloud and carrier infrastructure. In *Proc. European Workshop on Software Defined Networking (EWSN)*, 2014.
- [368] Michael Borokhovich, Liron Schiff, and Stefan Schmid. Provable data plane connectivity with local fast failover: Introducing openflow graph algorithms. In *Proc. ACM SIGCOMM Workshop on Hot Topics in Software Defined Networking (HotSDN)*, 2014.
- [369] Julius Schulz-Zander, Nadi Sarrar, and Stefan Schmid. Toward a scalable and near-sighted control plane architecture for wifi sdns (poster). In *Proc. ACM SIGCOMM Workshop on Hot Topics in Software Defined Networking (HotSDN)*, 2014.
- [370] Matthieu Roy, Stefan Schmid, and Gilles Tredan. Modeling and measuring graph similarity: The case for centrality distance. In *Proc. 10th ACM International Workshop on Foundations of Mobile Computing (FOMC)*, 2014.
- [371] Yvonne Anne Pignolet, Gilles Tredan, and Stefan Schmid. Adversarial topology discovery in network virtualization environments: A threat for isps? *Distributed Computing (DIST)*, 28, 2015.
- [372] Marcin Bienkowski, Nadi Sarrar, Stefan Schmid, and Steve Uhlig. Competitive fib aggregation without update churn. In *Proc. 34th International Conference on Distributed Computing Systems (ICDCS)*, 2014.
- [373] Dan Levin, Marco Canini, Stefan Schmid, Fabian Schaffert, and Anja Feldmann. Panopticon: Reaping the benefits of incremental sdn deployment in enterprise networks. In *Proc. USENIX Annual Technical Conference (ATC)*, 2014.
- [374] Galen Wilkerson, Ramin Khalili, and Stefan Schmid. Urban mobility scaling: Lessons from little data. In *Proc. 6th IEEE International Workshop on Network Science for Communication Networks (NetSciCom)*, 2014.

- [375] Marco Canini, Daniele De Cicco, Petr Kuznetsov, Dan Levin, Stefan Schmid, and Stefano Vissicchio. STN: A robust and distributed sdn control plane. In *Proc. Open Networking Summit (ONS)*, 2014.
- [376] Julius Schulz-Zander, Nadi Sarrar, and Stefan Schmid. Aeroflux: A near-sighted controller architecture for software-defined wireless networks. In *Proc. Open Networking Summit (ONS)*, 2014.
- [377] Vasileios Kotronis, Xenofontas Dimitropoulos, Rowan Klöti, Bernhard Ager, Panagiotis Georgopoulos, and Stefan Schmid. Control exchange points: Providing qos-enabled end-to-end services via sdn-based inter-domain routing orchestration. In *Proc. Open Networking Summit (ONS)*, 2014.
- [378] Dominic Meier, Yvonne Anne Oswald, Stefan Schmid, and Roger Wattenhofer. On the windfall and price of friendship: Inoculation strategies on social networks. *Elsevier Computer Networks (COMNET)*, 2014.
- [379] Matthias Rost, Stefan Schmid, and Anja Feldmann. It’s about time: On optimal virtual network embeddings under temporal flexibilities. In *Proc. IEEE International Parallel and Distributed Processing Symposium (IPDPS)*, May 2014.
- [380] Adrian Ogierman, Andrea Richa, Christian Scheideler, Stefan Schmid, and Jin Zhang. Competitive mac under adversarial sinr. In *Proc. IEEE INFOCOM*, April 2014.
- [381] Chen Avin, Omer Dunay, and Stefan Schmid. Strategies for traffic-aware vm migration (poster paper). In *Proc. 6th IEEE/ACM International Conference on Utility and Cloud Computing (UCC)*, December 2013.
- [382] Matthias Rost and Stefan Schmid. Virtucast: Multicast and aggregation with in-network processing (an exact single-commodity algorithm). In *Proc. 17th International Conference on Principles of Distributed Systems (OPODIS)*, December 2013.
- [383] Michael Borokhovich and Stefan Schmid. How (not) to shoot in your foot with sdn local fast failover: A load-connectivity tradeoff. In *Proc. 17th International Conference on Principles of Distributed Systems (OPODIS)*, December 2013.
- [384] Dominik Gall, Riko Jacob, Andrea Richa, Christian Scheideler, Stefan Schmid, and Hanjo Taeubig. A note on the parallel runtime of self-stabilizing graph linearization. *Theory of Computing Systems (TOCS)*, 2014.
- [385] Carlo Fuerst, Stefan Schmid, and Anja Feldmann. Virtual network embedding with collocation: Benefits and limitations of pre-clustering. In *Proc. 2nd IEEE International Conference on Cloud Networking (CLOUDNET)*, November 2013.
- [386] Marco Canini, Petr Kuznetsov, Dan Levin, and Stefan Schmid. Brief announcement: Towards distributed and reliable software defined networking. In *Proc. 27th International Symposium on Distributed Computing (DISC)*, October 2013.
- [387] Marcin Bienkowski, Nadi Sarrar, Stefan Schmid, and Steve Uhlig. Brief announcement: Dynamic forwarding table aggregation without update churn: The case of dependent prefixes. In *Proc. 27th International Symposium on Distributed Computing (DISC)*, October 2013.
- [388] Arne Ludwig, Stefan Schmid, and Anja Feldmann. Specificity vs. flexibility: On the embedding cost of a virtual network. In *Proc. 25th International Teletraffic Congress (ITC)*, September 2013.
- [389] Chen Avin, Michael Borokhovich, and Stefan Schmid. Obst: A self-adjusting peer-to-peer overlay based on multiple bst. In *Proc. 13th IEEE International Conference on Peer-to-Peer Computing (P2P)*, September 2013.

- [390] Ishan Vaishnavi, David Perez-Caparrós, Stefan Schmid, and Ashiq Khan. An architecture for creating and managing virtual networks. In *Proc. 24th IEEE International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)*, September 2013.
- [391] Johannes Grassler, Gregor Schaffrath, and Stefan Schmid. A federated cloudnet architecture: The pip and the vnp role. *Information Technology, Oldenbourg Verlag*, 2013.
- [392] Stefan Schmid and Jukka Suomela. Exploiting locality in distributed sdn control. In *Proc. ACM SIGCOMM Workshop on Hot Topics in Software Defined Networking (HotSDN)*, August 2013.
- [393] Marco Canini, Petr Kuznetsov, Dan Levin, and Stefan Schmid. Software transactional networking: Concurrent and consistent policy composition. In *Proc. ACM SIGCOMM Workshop on Hot Topics in Software Defined Networking (HotSDN)*, August 2013.
- [394] Marcin Bienkowski and Stefan Schmid. Competitive fib aggregation: Online ski rental on the trie. In *Proc. 20th International Colloquium on Structural Information and Communication Complexity (SIROCCO)*, July 2013.
- [395] Baruch Awerbuch, Andrea Richa, Christian Scheideler, Stefan Schmid, and Jin Zhang. Principles of robust medium access and an application to leader election. *ACM Transactions on Algorithms (TALG)*, 2013.
- [396] Yvonne Anne Pignolet, Stefan Schmid, and Gilles Tredan. Request complexity of vnet topology extraction: Dictionary-based attacks. In *International Conference on Networked Systems (NETYS)*, May 2013.
- [397] Marcin Bienkowski, Anja Feldmann, Johannes Grassler, Gregor Schaffrath, and Stefan Schmid. The wide-area virtual service migration problem: A competitive analysis approach. *IEEE/ACM Transactions on Networking (ToN)*, 2014.
- [398] Yvonne-Anne Pignolet, Stefan Schmid, and Gilles Tredan. Misleading stars: What cannot be measured in the internet? *Distributed Computing (DIST)*, 2013.
- [399] Chen Avin, Bernhard Haeupler, Zvi Lotker, Christian Scheideler, and Stefan Schmid. Locally self-adjusting tree networks. In *27th IEEE International Parallel and Distributed Processing Symposium (IPDPS)*, May 2013.
- [400] Dan Levin, Marco Canini, Stefan Schmid, and Anja Feldmann. Toward transitional sdn deployment in enterprise networks. In *Proc. Open Networking Summit (ONS)*, Apr 2013.
- [401] Johannes Grassler, Anja Feldmann, and Stefan Schmid. Move-with-the-sun or move-with-the-moon? wide-area cloudnet migrations under latency and resource constraints. In *Demo at 32nd IEEE Conference on Computer Communications (INFOCOM)*, April 2013.
- [402] Johannes Schneider and Stefan Schmid. Optimal bounds for online page migration with generalized migration costs. In *32nd IEEE Conference on Computer Communications (INFOCOM)*, April 2013.
- [403] Yvonne-Anne Pignolet, Gilles Tredan, and Stefan Schmid. Adversarial vnet embeddings: A threat for isps? In *INFOCOM Mini-Conference*, April 2013.
- [404] Guy Even, Moti Medina, Gregor Schaffrath, and Stefan Schmid. Competitive and deterministic embeddings of virtual networks. *Elsevier Theoretical Computer Science (TCS)*, 2013.
- [405] Riko Jacob, Stephan Ritscher, Christian Scheideler, and Stefan Schmid. Towards higher-dimensional topological self-stabilization: A distributed algorithm for delaunay graphs. *Elsevier Theoretical Computer Science (TCS)*, 457, 2012.

- [406] Xinhui Hu, Stefan Schmid, Andrea Richa, and Anja Feldmann. Optimal migration contracts in virtual networks: Pay-as-you-come vs pay-as-you-go pricing. In *Proc. 14th International Conference on Distributed Computing and Networking (ICDCN)*, 2013.
- [407] Carlo Fuerst, Stefan Schmid, and Anja Feldmann. On the benefit of collocation in virtual network embeddings (short paper). In *Proc. 1st IEEE International Conference on Cloud Networking (CLOUD-NET)*, 2012.
- [408] Gregor Schaffrath, Stefan Schmid, and Anja Feldmann. Optimizing long-lived cloudnets with migrations. In *Proc. 5th IEEE/ACM International Conference on Utility and Cloud Computing (UCC)*, 2012.
- [409] Arne Ludwig, Stefan Schmid, and Anja Feldmann. The price of specificity in the age of network virtualization (short paper). In *Proc. 5th IEEE/ACM International Conference on Utility and Cloud Computing (UCC)*, 2012.
- [410] Stefan Schmid, Chen Avin, Christian Scheideler, and Bernhard Haeupler. Brief announcement: Splaynets (towards self-adjusting distributed data structures). In *Proc. 26th International Symposium on Distributed Computing (DISC)*, 2012.
- [411] Yvonne Anne Pignolet, Stefan Schmid, and Gilles Tredan. Brief announcement: Do vnet embeddings reveal isp topology? In *Proc. 26th International Symposium on Distributed Computing (DISC)*, 2012.
- [412] Raphael Eidenbenz, Thomas Locher, Stefan Schmid, and Roger Wattenhofer. Boosting market liquidity of peer-to-peer systems through cyclic trading. In *Proc. 12th IEEE International Conference on Peer-to-Peer Computing (P2P)*, 2012.
- [413] Andrea Richa, Christian Scheideler, Stefan Schmid, and Jin Zhang. An efficient and fair mac protocol robust to reactive interference. *IEEE/ACM Transactions on Networking (ToN)*, 2013.
- [414] Andrea Richa, Christian Scheideler, Stefan Schmid, and Jin Zhang. An efficient and fair mac protocol robust to reactive interference. *Distributed Computing (DIST)*, 2013.
- [415] Gregor Schaffrath, Stefan Schmid, Ishan Vaishnavi, Ashiq Khan, and Anja Feldmann. A resource description language with vagueness support for multi-provider cloud networks. In *Proc. International Conference on Computer Communication Networks (ICCCN)*, 2012.
- [416] Andréa Richa, Christian Scheideler, Stefan Schmid, and Jin Zhang. Competitive and fair throughput for co-existing networks under adversarial interference. In *Proc. 31st Annual ACM Symposium on Principles of Distributed Computing (PODC)*, 2012.
- [417] Doris Schioeberg, Fabian Schneider, Harald Schioeberg, Stefan Schmid, Steve Uhlig, and Anja Feldmann. Tracing the birth of an osn: Social graph and profile analysis in google+. In *ACM Web Science (WebSci)*, 2012.
- [418] Nadi Sarrar, Stefan Schmid, Steve Uhlig, and Anja Feldmann. Poster: Impact of fib aggregation on traffic offloading. In *9th USENIX Symposium on Networked Systems Design and Implementation (NSDI)*, 2012.
- [419] Anja Feldmann, Gregor Schaffrath, and Stefan Schmid. Cloudnets: Combining clouds with networking. *ERCIM News*, (88):5657, January 2012.
- [420] Guy Even, Moti Medina, Gregor Schaffrath, and Stefan Schmid. Competitive and deterministic embeddings of virtual networks. In *13th International Conference on Distributed Computing and Networking (ICDCN)*, 2012.

- [421] Yvonne Anne Pignolet, Gilles Trédan, and Stefan Schmid. Misleading stars: What cannot be measured in the internet? In David Peleg, editor, *Proc. 25th International Symposium on Distributed Computing (DISC)*, 2011.
- [422] Raphael Eidenbenz, Yvonne Anne Pignolet, Stefan Schmid, and Roger Wattenhofer. Cost and complexity of harnessing games with payments. *International Game Theory Review*, 13(1), 2011.
- [423] Thomas Moscibroda, Stefan Schmid, and Roger Wattenhofer. Topological implications of selfish neighbor selection in unstructured peer-to-peer networks. *Algorithmica*, 61(2):419–446, 2011.
- [424] Andréa Richa, Christian Scheideler, Stefan Schmid, and Jin Zhang. Towards jamming-resistant and competitive medium access in the sinr model. In *Proc. 3rd Annual ACM S3 Workshop*, 2011.
- [425] Dushyant Arora, Marcin Bienkowski, Anja Feldmann, Gregor Schaffrath, and Stefan Schmid. Online strategies for intra and inter provider service migration in virtual networks. In *Proc. Principles, Systems and Applications of IP Telecommunications (IPTComm)*, 2011.
- [426] Thomas Locher, Stefan Schmid, and Roger Wattenhofer. edonkey and emule’s kad: Measurements and attacks. *Fundamenta Informaticae*, 109(4):383–403, 2011.
- [427] Andréa Richa, Christian Scheideler, Stefan Schmid, and Jin Zhang. Competitive and fair medium access despite reactive jamming. In *Proc. 31st International Conference on Distributed Computing Systems (ICDCS)*, New York, NY, USA, June 2011. IEEE ComSoc, IEEE.
- [428] Andréa Richa, Christian Scheideler, Stefan Schmid, and Jin Zhang. Self-stabilizing leader election for single-hop wireless networks despite jamming. In *Proc. 12th ACM International Symposium on Mobile Ad Hoc Networking and Computing (MobiHoc)*, 2011.
- [429] Dushyant Arora, Anja Feldmann, Gregor Schaffrath, and Stefan Schmid. On the benefit of virtualization: Strategies for flexible server allocation. In *Proc. USENIX Workshop on Hot Topics in Management of Internet, Cloud, and Enterprise Networks and Services (Hot-ICE)*, 2011.
- [430] Marcin Bienkowski, Anja Feldmann, Dan Jurca, Wolfgang Kellerer, Gregor Schaffrath, Stefan Schmid, and Jörg Widmer. Competitive analysis for service migration in vnets. In *Proc. 2nd ACM SIGCOMM Workshop on Virtualized Infrastructure Systems and Architectures (VISA)*, pages 17–24, 2010.
- [431] Andréa Richa, Christian Scheideler, Stefan Schmid, and Jin Zhang. A jamming-resistant mac protocol for multi-hop wireless networks. In *24th International Symposium on Distributed Computing (DISC)*, volume 6343 of *Lecture Notes in Computer Science (LNCS)*, pages 179–193, Berlin / Heidelberg, Germany, 2010.
- [432] Stefan Schmid. Robust architectures for open distributed systems and topological self-stabilization (invited paper). In *Proc. 3rd ACM SIGOPS/SIGACT Workshop on Reliability, Availability, and Security (WRAS)*, July 2010. Invited Paper.
- [433] Andréa Richa, Christian Scheideler, Stefan Schmid, and Jin Zhang. Brief announcement: Towards robust medium access in multi-hop networks. In *Proc. 29th Annual ACM SIGACT-SIGOPS Symposium on Principles of Distributed Computing (PODC 2010)*, pages 114–115, New York, NY, USA, July 2010. ACM. Brief Announcement.
- [434] Marcin Bienkowski and Stefan Schmid. Online function tracking with generalized penalties. In *Proc. 12th Scandinavian Symposium and Workshops on Algorithm Theory (SWAT 2010)*, volume 6139 of *Lecture Notes in Computer Science (LNCS)*, pages 359–370, Berlin / Heidelberg, Germany, June 2010. Springer.

- [435] Petr Kuznetsov and Stefan Schmid. Towards network games with social preferences. In Boaz Patt-Schamir and Tinaz Ekim, editors, *Proc. 17th International Colloquium on Structural Information and Communication Complexity (SIROCCO)*, number 6058 in Lecture Notes of Computer Science, pages 14–28, Berlin, Heidelberg, New York, June 2010. Springer. See also arXiv TechReport No. 1001.5134.
- [436] Marcin Bienkowski, Leszek Gasieniec, Marek Klonowski, Mirosław Korzeniowski, and Stefan Schmid. Event extent estimation. In Boaz Patt-Schamir and Tinaz Ekim, editors, *Proc. 17th International Colloquium on Structural Information and Communication Complexity (SIROCCO)*, number 6058 in Lecture Notes in Computer Science (LNCS), pages 57–71, June 2010.
- [437] Fabian Kuhn, Stefan Schmid, and Roger Wattenhofer. Towards worst-case churn resistant peer-to-peer systems. *Distributed Computing Journal (DC)*, 22(4):249–267, May 2010.
- [438] Yvonne Anne Pignolet, Stefan Schmid, and Roger Wattenhofer. Tight bounds for delay-sensitive aggregation. *Discrete Mathematics and Theoretical Computer Science Journal*, 12(1):39–58, 2010.
- [439] Dominik Gall, Riko Jacob, Andréa Richa, Christian Scheideler, Stefan Schmid, and Hanjo Täubig. Time complexity of distributed topological self-stabilization: The case of graph linearization. In *Proc. 9th Latin American Theoretical Informatics Symposium (LATIN)*, volume 6034 of *Lecture Notes in Computer Science (LNCS)*, pages 294–305, Berlin / Heidelberg, Germany, April 2010. Springer.
- [440] Thomas Locher, David Mysicka, Stefan Schmid, and Roger Wattenhofer. Poisoning the kad network. In *11th International Conference on Distributed Computing and Networking (ICDCN)*, volume 5935 of *Lecture Notes in Computer Science (LNCS)*, pages 195–206, Berlin / Heidelberg, Germany, January 2010. Springer.
- [441] Thomas Moscibroda, Stefan Schmid, and Roger Wattenhofer. The price of malice: A game-theoretic framework for malicious behavior in distributed systems. *Internet Mathematics (IM)*, 6(2):125–155, 2009.
- [442] Riko Jacob, Stephan Ritscher, Christian Scheideler, and Stefan Schmid. A self-stabilizing and local delaunay graph construction. In *Algorithms and Computations: Proc. 20th International Symposium on Algorithms and Computation (ISAAC)*, volume 5878 of *Lecture Notes in Computer Science (LNCS)*, Berlin / Heidelberg, Germany, December 2009. Springer.
- [443] Dominik Gall, Riko Jacob, Andréa Richa, Christian Scheideler, Stefan Schmid, and Hanjo Täubig. Brief announcement: On the time complexity of distributed topological self-stabilization. In *Stabilization, Safety, and Security of Distributed Systems – Proc. 11th International Symposium on Stabilization, Safety, and Security of Distributed Systems (SSS)*, volume 5873 of *Lecture Notes in Computer Science (LNCS)*, pages 781–782, Berlin / Heidelberg, Germany, November 2009. Springer. Brief Announcement.
- [444] Thomas Locher, David Mysicka, Stefan Schmid, and Roger Wattenhofer. A peer activity study in edonkey and kad. In *International Workshop on Dynamic Networks: Algorithms and Security (DYNAS)*, September 2009. Invited Paper.
- [445] Matthias Baumgart, Christian Scheideler, and Stefan Schmid. A dos-resilient information system for dynamic data management. In *21st ACM Symposium on Parallelism in Algorithms and Architectures (SPAA)*, pages 300–309, August 2009.
- [446] Riko Jacob, Andréa Richa, Christian Scheideler, Stefan Schmid, and Hanjo Täubig. A polylogarithmic time algorithm for distributed self-stabilizing skip graphs. In *28th ACM Symposium on Principles of Distributed Computing (PODC)*, pages 131–140, August 2009.
- [447] Christian Scheideler and Stefan Schmid. A distributed and oblivious heap. In *36th International Colloquium on Automata, Languages and Programming (ICALP)*, volume 5556 of *Lecture Notes in Computer Science (LNCS)*, pages 571–582, Berlin / Heidelberg, Germany, July 2009. Springer.

- [448] Dominic Meier, Yvonne Anne Pignolet, Stefan Schmid, and Roger Wattenhofer. Speed dating despite jammers. In *5th IEEE International Conference on Distributed Computing in Sensor Systems (DCOSS)*, volume 5516 of *Lecture Notes in Computer Science (LNCS)*, pages 1–14, Berlin / Heidelberg, Germany, June 2009. Springer.
- [449] Thomas Locher, Remo Meier, Roger Wattenhofer, and Stefan Schmid. Robust live media streaming in swarms. In *19th International Workshop on Network and Operating Systems Support for Digital Audio and Video (NOSSDAV)*, pages 121–126, June 2009.
- [450] Raphael Eidenbenz and Stefan Schmid. Combinatorial agency with audits. In *IEEE International Conference on Game Theory for Networks (GameNets)*, pages 374–383, May 2009.
- [451] Thomas Moscibroda and Stefan Schmid. On mechanism design without payments for throughput maximization. In *28th IEEE Conference on Computer Communications (INFOCOM)*, pages 972–980, April 2009.
- [452] Fabian Kuhn, Thomas Locher, and Stefan Schmid. Distributed computation of the mode. In *27th ACM Symposium on Principles of Distributed Computing (PODC)*, pages 15–24, August 2008.
- [453] Yvonne Anne Oswald, Stefan Schmid, and Roger Wattenhofer. Tight bounds for delay-sensitive aggregation. In *27th ACM Symposium on Principles of Distributed Computing (PODC)*, pages 195–202, August 2008.
- [454] Dominic Meier, Yvonne Anne Oswald, Stefan Schmid, and Roger Wattenhofer. On the windfall of friendship: Inoculation strategies on social networks. In *9th ACM Conference on Electronic Commerce (EC)*, pages 294–301, July 2008.
- [455] Bernard Mans, Stefan Schmid, and Roger Wattenhofer. Distributed disaster disclosure. In Joachim Gudmundsson, editor, *11th Scandinavian Workshop on Algorithm Theory (SWAT)*, volume 5124 of *Lecture Notes in Computer Science (LNCS)*, pages 246–257, Berlin / Heidelberg, Germany, July 2008. Springer.
- [456] Stefan Schmid and Roger Wattenhofer. Structuring unstructured peer-to-peer networks. In *14th Annual IEEE International Conference on High Performance Computing (HiPC)*, volume 4873 of *Lecture Notes in Computer Science (LNCS)*, pages 432–442, Berlin / Heidelberg, Germany, December 2007. Springer.
- [457] Raphael Eidenbenz, Yvonne Anne Oswald, Stefan Schmid, and Roger Wattenhofer. Manipulation in games. In *18th International Symposium on Algorithms and Computation (ISAAC)*, volume 4835 of *Lecture Notes in Computer Science (LNCS)*, pages 365–376, Berlin / Heidelberg, Germany, December 2007. Springer.
- [458] Thomas Locher, Remo Meier, Stefan Schmid, and Roger Wattenhofer. Push-to-pull peer-to-peer live streaming. In *21st International Symposium on Distributed Computing (DISC)*, volume 4731 of *Lecture Notes in Computer Science (LNCS)*, pages 388–402, Berlin / Heidelberg, Germany, September 2007. Springer.
- [459] Thomas Locher, Stefan Schmid, and Roger Wattenhofer. Rescuing tit-for-tat with source coding. In *7th IEEE International Conference on Peer-to-Peer Computing (P2P)*, pages 3–10, September 2007.
- [460] Raphael Eidenbenz, Yvonne Anne Oswald, Stefan Schmid, and Roger Wattenhofer. Mechanism design by creditability. In *1st International Conference on Combinatorial Optimization and Applications (COCOA)*, volume 4616 of *Lecture Notes in Computer Science (LNCS)*, pages 208–219, Berlin / Heidelberg, Germany, August 2007. Springer.

- [461] Stefan Schmid and Roger Wattenhofer. Dynamic internet congestion with bursts. In Yves Robert, Manish Parashar, Ramamurthy Badrinath, and Viktor K. Prasanna, editors, *13th Annual IEEE International Conference on High Performance Computing (HiPC)*, volume 4297 of *Lecture Notes in Computer Science (LNCS)*, pages 159–170, Berlin / Heidelberg, Germany, December 2006. Springer.
- [462] Thomas Locher, Patrick Moor, Stefan Schmid, and Roger Wattenhofer. Free riding in bittorrent is cheap. In *5th Workshop on Hot Topics in Networks (HotNets)*, pages 85–90, November 2006.
- [463] Dominik Grolimund, Luzius Meisser, Stefan Schmid, and Roger Wattenhofer. Cryptree: A folder tree structure for cryptographic file systems. In *25th IEEE Symposium on Reliable Distributed Systems (SRDS)*, pages 189–198, October 2006.
- [464] Stefan Schmid and Roger Wattenhofer. A tcp with guaranteed performance in networks with dynamic congestion and random wireless losses. In *2nd Annual International Wireless Internet Conference (WICON)*, August 2006.
- [465] Thomas Moscibroda, Stefan Schmid, and Roger Wattenhofer. On the topologies formed by selfish peers. In *25th Annual Symposium on Principles of Distributed Computing (PODC)*, pages 133–142, July 2006.
- [466] Thomas Moscibroda, Stefan Schmid, and Roger Wattenhofer. When selfish meets evil: Byzantine players in a virus inoculation game. In *25th Annual Symposium on Principles of Distributed Computing (PODC)*, pages 35–44, July 2006.
- [467] Fabian Kuhn, Stefan Schmid, Joest Smit, and Roger Wattenhofer. A blueprint for constructing peer-to-peer systems robust to dynamic worst-case joins and leaves. In *14th IEEE International Workshop on Quality of Service (IWQoS)*, pages 12–19, June 2006.
- [468] Dominik Grolimund, Luzius Meisser, Stefan Schmid, and Roger Wattenhofer. Havelaar: A robust and efficient reputation system for active peer-to-peer systems. In *1st Workshop on the Economics of Networked Systems (NetEcon)*, pages 69–74, June 2006.
- [469] Stefan Schmid and Roger Wattenhofer. Algorithmic models for sensor networks. In *14th International Workshop on Parallel and Distributed Real-Time Systems (WPDRTS), in conjunction with IPDPS*, pages 1–11, April 2006. Invited Paper.
- [470] Thomas Moscibroda, Stefan Schmid, and Roger Wattenhofer. On the topologies formed by selfish peers. In *5th International Workshop on Peer-to-Peer Systems (IPTPS)*, pages 133–142, February 2006.
- [471] Fabian Kuhn, Stefan Schmid, and Roger Wattenhofer. A self-repairing peer-to-peer system resilient to dynamic adversarial churn. In *4th International Workshop on Peer-To-Peer Systems (IPTPS)*, volume 3640 of *Lecture Notes in Computer Science (LNCS)*, pages 13–23, Berlin / Heidelberg, Germany, 2005. Springer.
- [472] Pascal von Rickenbach, Stefan Schmid, Roger Wattenhofer, and Aaron Zollinger. A robust interference model for wireless ad-hoc networks. In *5th International Workshop on Algorithms for Wireless, Mobile, Ad Hoc and Sensor Networks (WMAN)*, pages 1–8, April 2005.